

D09 Human-Facing GPIM, Capital, Output, and Distribution Visual Validation Report

Capacity Utilization US-Chile Source-of-Truth Pipeline

2026-07-01

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1 Purpose and Status of D09

D09 is a human-facing visual validation layer. All indices, ratios, shares, and residuals produced here are REPORT ONLY, INSPECTION ONLY, NOT SOURCE OF TRUTH, NOT MODEL READY, and require D10 authorization for transformation use.

2 Locked Source-of-Truth and GPIM Repair Lineage

stage	decision	role
D05	AUTHORIZE_D06.GPIM_REFREEZE_WITH_GPIM_GUARDED_pKN	pKN guardian
D06	AUTHORIZE_D07.CAPACITY_PANEL_CONSUMPTION	GPIM refreeze
D07-0	AUTHORIZE_D07.LEVEL_ACCOUNTING_PANEL_CONSUMPTION	consumption contract
D07	AUTHORIZE_D08.SOURCE_OF_TRUTH_REVIEW	level/accounting panel
D08	AUTHORIZE_D09.TRANSFORMATION_PLANNING	audit and repair-regression r
D09	visual validation report	human inspection interface

3 GPIM Equations and Capital-Stock Architecture

object	equation	status
Real investment	$I^R_{j,t} = I^C_{j,t} / (p^K_{j,t} / 100)$	locked D05-D08 lineage
Gross surviving GPIM stock	$K^R_{j,t} = \sum_{a=0}^A s_j(a) I^R_{j,t-a}$	locked D05-D08 lineage
Current-cost valuation	$K^C_{j,t} = K^R_{j,t} p^K_{j,t} / 100$	locked D05-D08 lineage
Real capacity aggregation	$K^R_{cap,t} = K^R_{ME,t} + K^R_{NRC,t}$	locked D05-D08 lineage
Current-cost capacity aggregation	$K^C_{cap,t} = K^C_{ME,t} + K^C_{NRC,t}$	locked D05-D08 lineage
Capacity pKN	$p^K_{cap,t} = 100 K^C_{cap,t} / K^R_{cap,t}$	locked D05-D08 lineage

$$I^R_{j,t} = \frac{I^C_{j,t}}{p^K_{j,t} / 100} \quad (1)$$

$$K^R_{j,t} = \sum_{a=0}^A s_j(a) I^R_{j,t-a} \quad (2)$$

$$K^C_{j,t} = K^R_{j,t} \frac{p^K_{j,t}}{100} \quad (3)$$

$$K^R_{cap,t} = K^R_{ME,t} + K^R_{NRC,t} \quad (4)$$

$$K^C_{cap,t} = K^C_{ME,t} + K^C_{NRC,t} \quad (5)$$

$$p^K_{cap,t} = 100 \times \frac{K^C_{cap,t}}{K^R_{cap,t}} \quad (6)$$

$$X_t^{index,b} = 100 \times \frac{X_t}{X_b} \quad (7)$$

$$\kappa_t = \frac{K^R_{cap,t}}{Y^R_{NFC,t}} \quad (8)$$

$$\omega^{GVA}_{NFC,t} = \frac{W_{NFC,t}}{GVA_{NFC,t}} \quad (9)$$

$$\omega^{NVA}_{NFC,t} = \frac{W_{NFC,t}}{NVA_{NFC,t}} \quad (10)$$

$$\pi^{GOS}_{NFC,t} = \frac{GOS_{NFC,t}}{GVA_{NFC,t}} \quad (11)$$

$$\pi^{NOS}_{NFC,t} = \frac{NOS_{NFC,t}}{NVA_{NFC,t}} \quad (12)$$

4 Real and Current-Cost GPIM Stock Levels

5 Visual Inspection Indices and Growth-Tendency Checks

Visual indices are D09 report-only inspection artifacts. They are not source-of-truth variables and are not authorized model transformations.

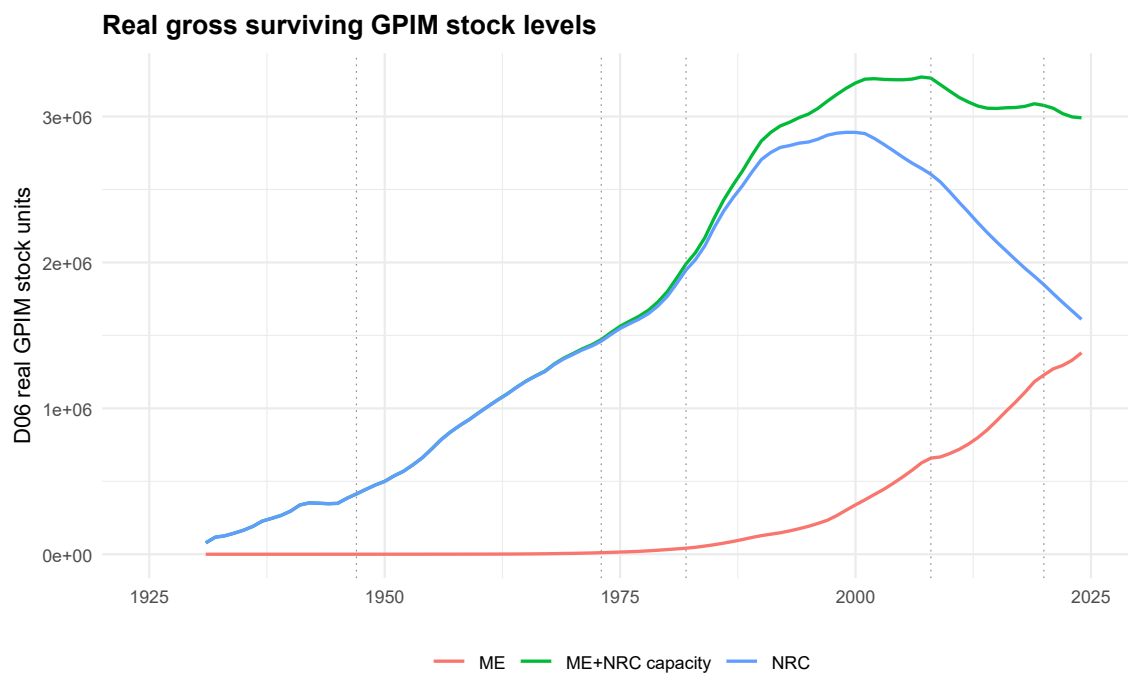


Figure 1: Real gross surviving GPIM stock levels.

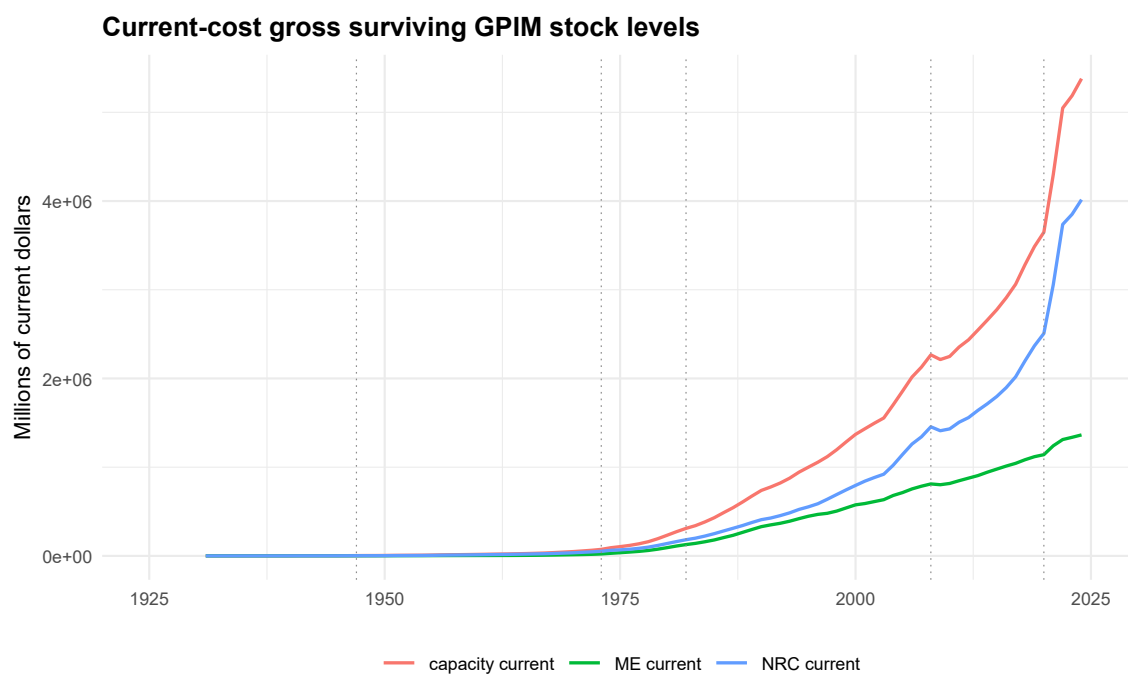


Figure 2: Current-cost gross surviving GPIM stock levels.

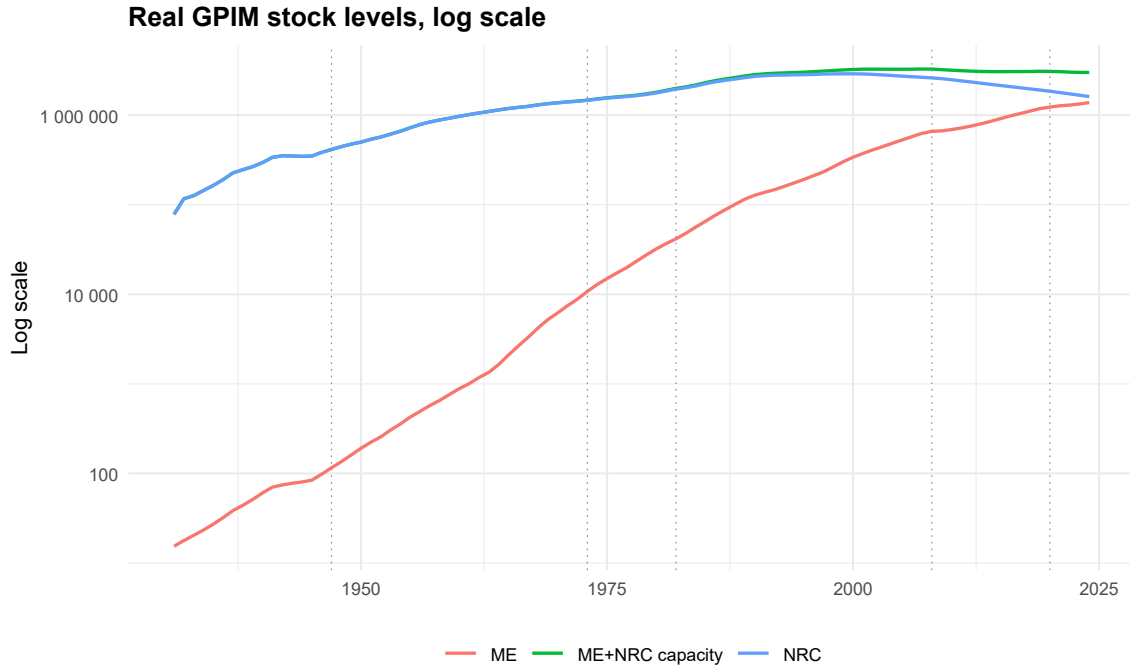


Figure 3: Real GPIM stock levels on log scale.

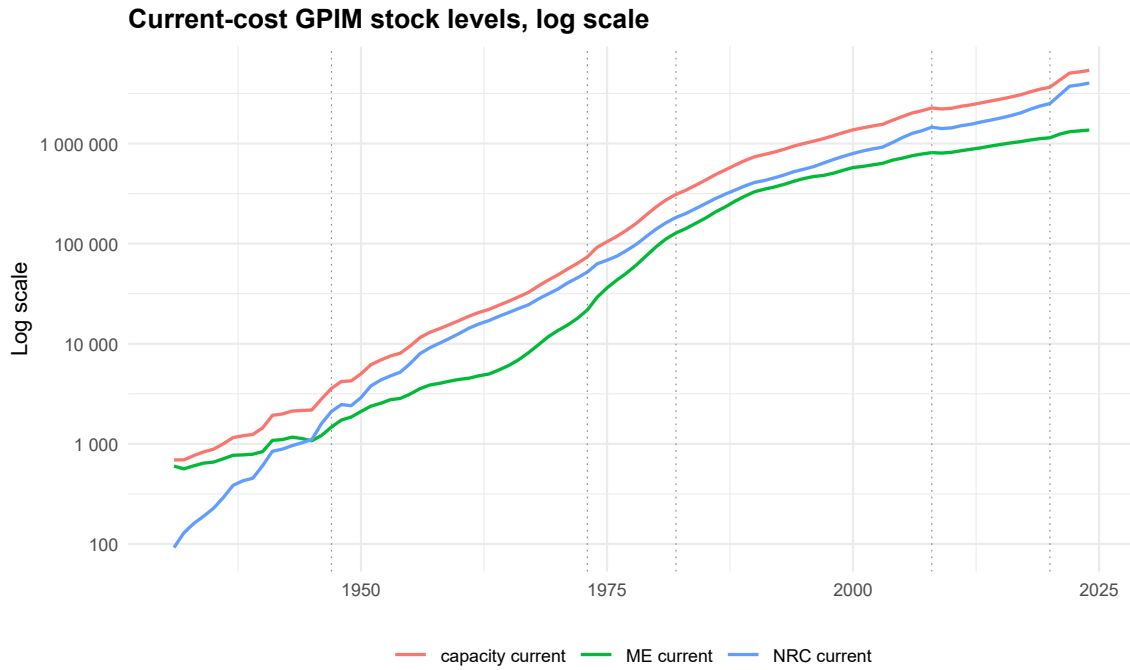


Figure 4: Current-cost GPIM stock levels on log scale.

6 Warmup, Survival Profiles, and NRC Review Flag

asset	warmup_length	survival_L	survival_alpha	warmup_to_L_ratio	sufficiency_status
ME	22	14	1.7	1.5714286	LIKELY_ADEQUATE
NRC	16	30	6 1.6	0.5333333	REVIEW_FLAG_SHORT_RELATIV

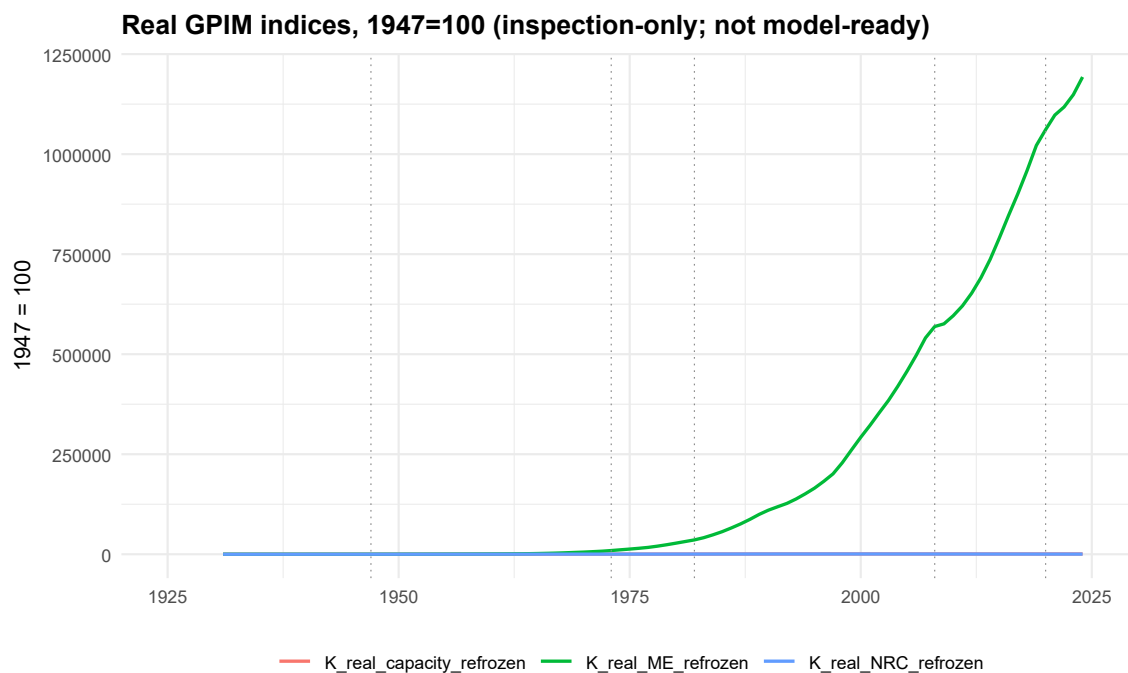


Figure 5: Real GPIM indices, 1947=100.

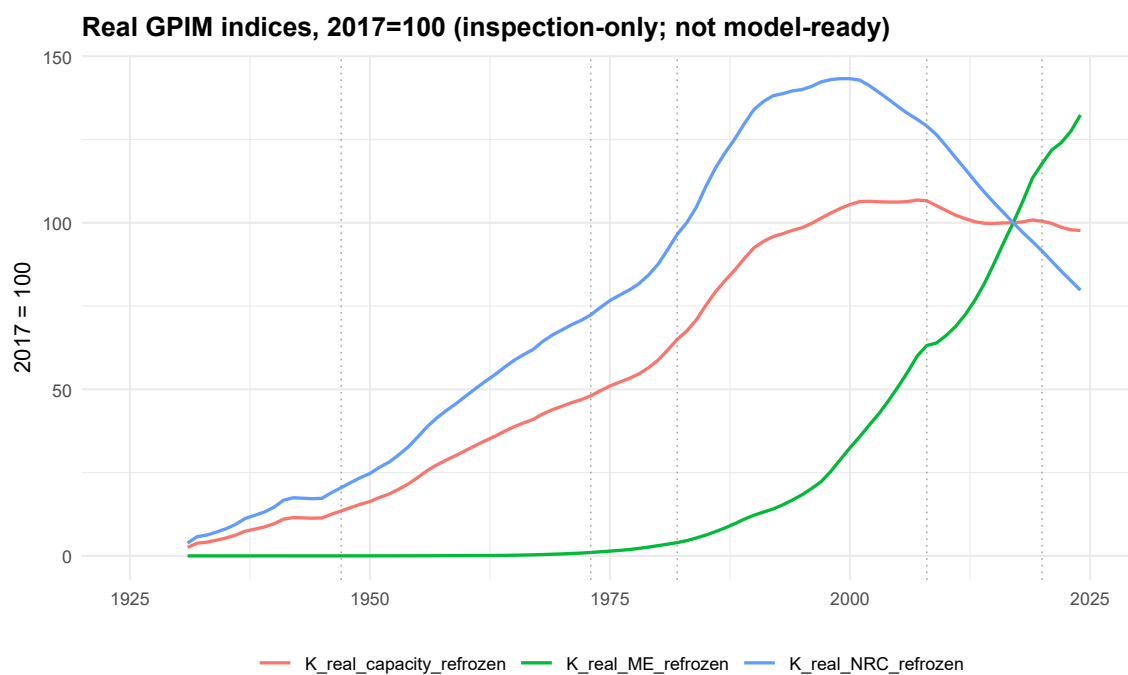


Figure 6: Real GPIM indices, 2017=100.

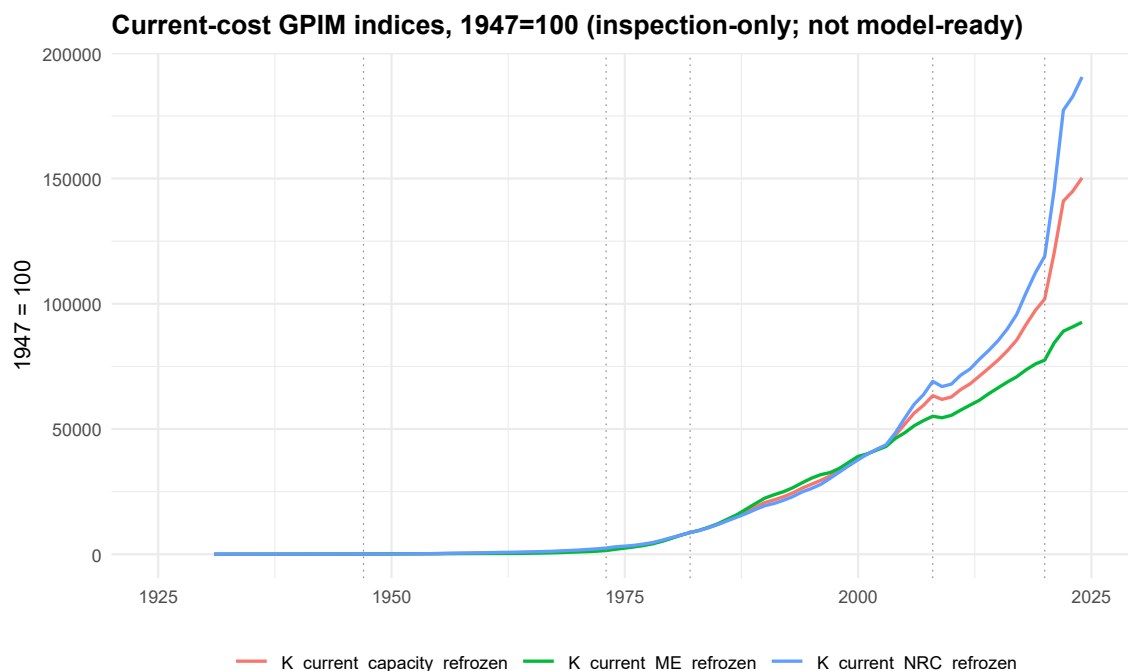


Figure 7: Current-cost GPIM indices, 1947=100.

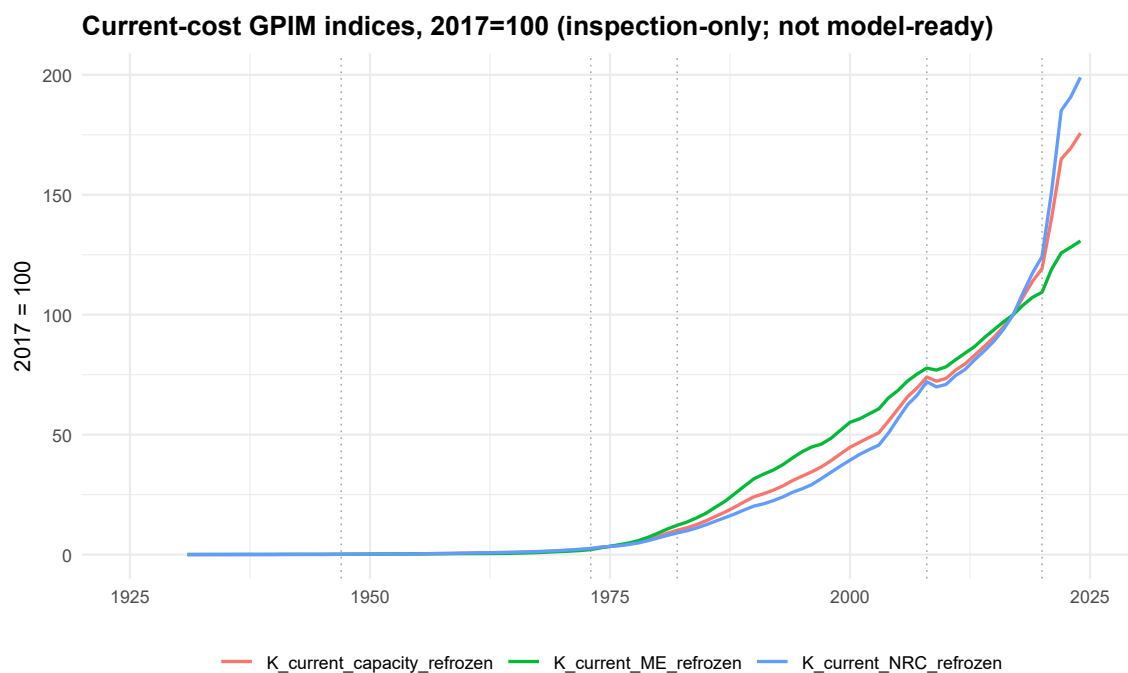


Figure 8: Current-cost GPIM indices, 2017=100.

7 Stock-Flow and Valuation Consistency Audits

object		metric	max_abs_residual	audit_status
ME		Lreal residual	0	PASS
NRC		Lreal residual	0	PASS
K_current_ME_refrozen_identity	8	current-cost identity	0	PASS
K_current_NRC_refrozen_identity		current-cost identity	0	PASS
K_current_capacity_refrozen_identity		current-cost identity	0	PASS

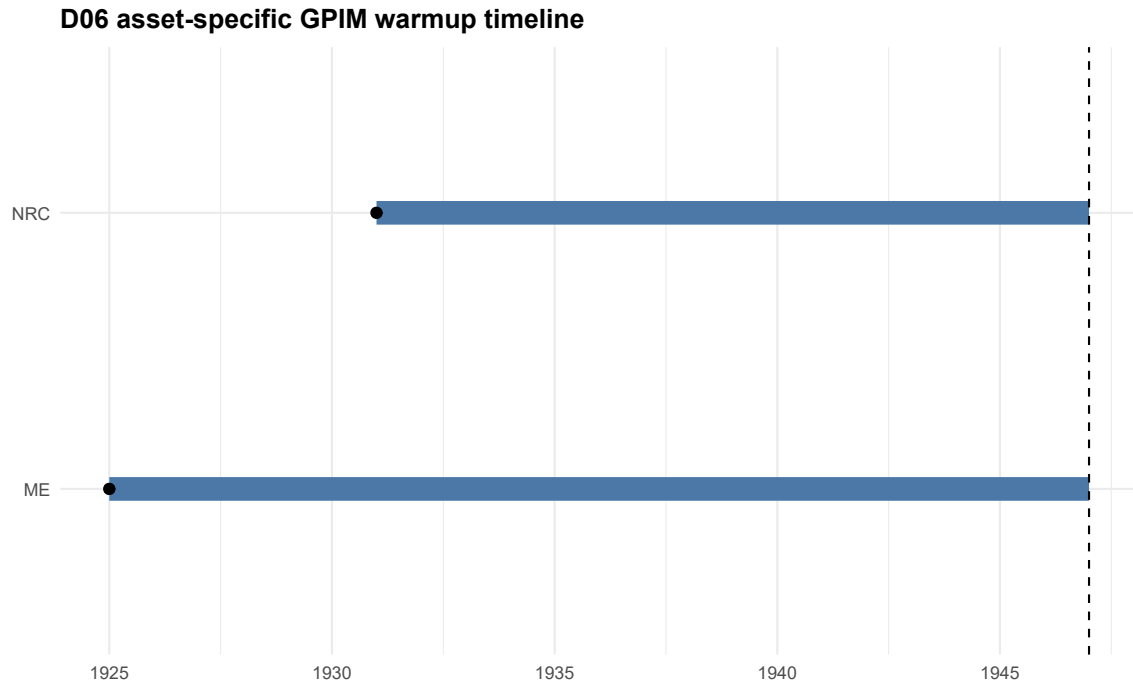


Figure 9: Asset-specific warmup timeline with analysis start.

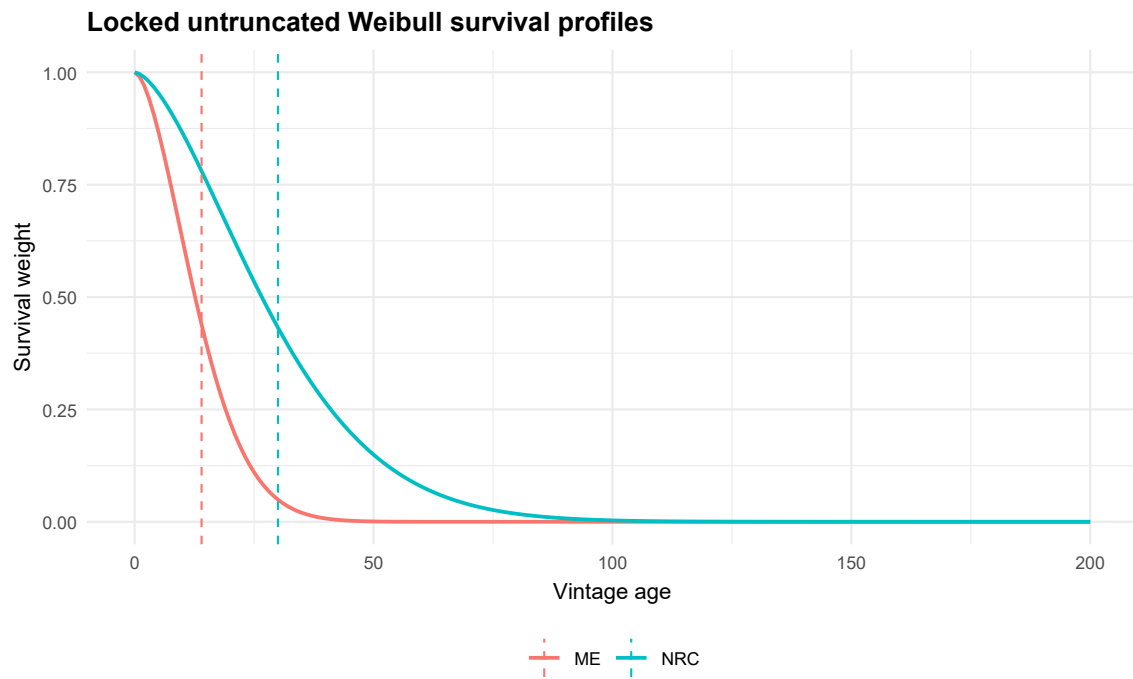


Figure 10: Locked survival profiles with service-life markers.

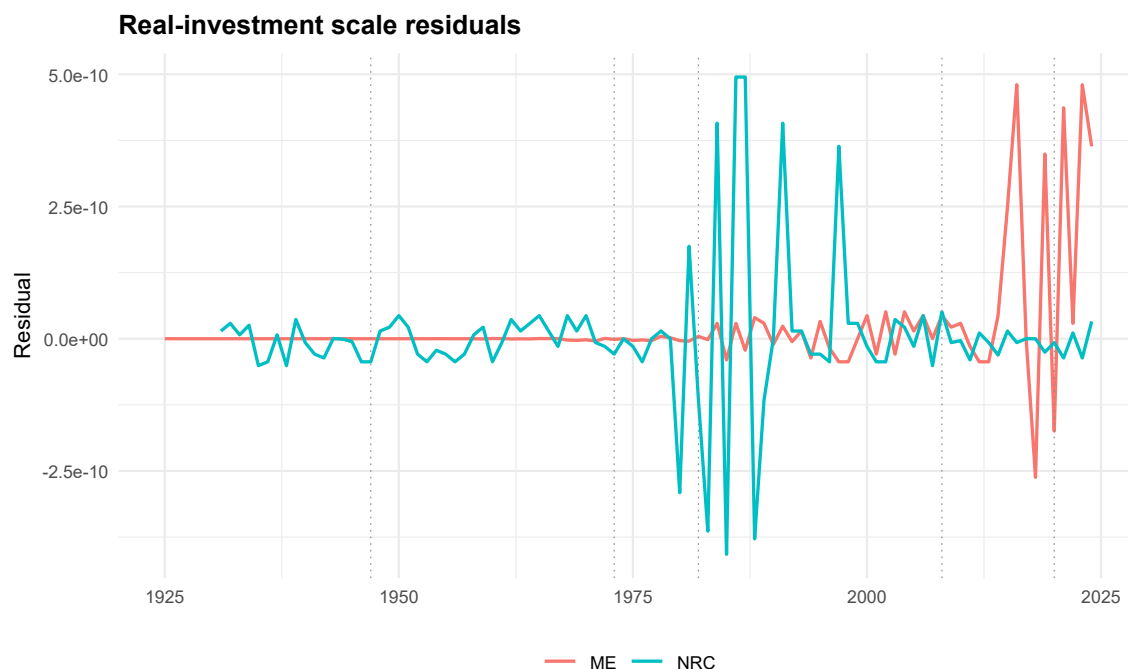


Figure 11: Real-investment scale residuals.

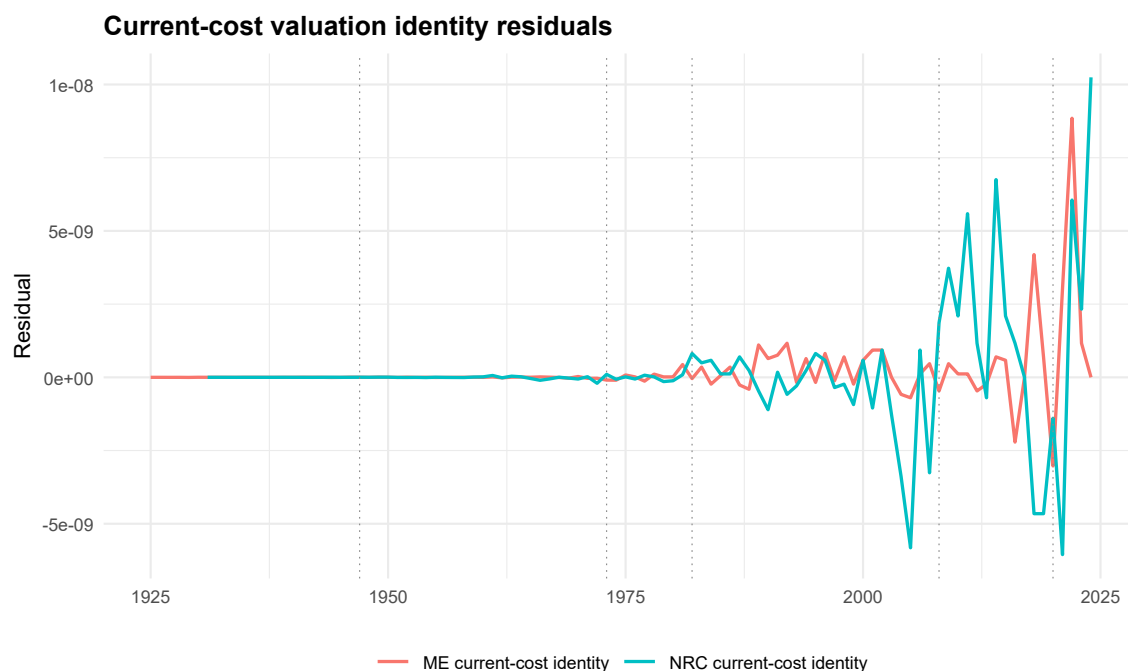


Figure 12: Current-cost valuation identity residuals.

8 ME/NRC Composition and Mechanization Diagnostics

9 Net-over-Gross Capital-Stock Comparisons

10 Alternative Capital Measures: Baseline, Comparison, Review-Only, Parked, and Excluded

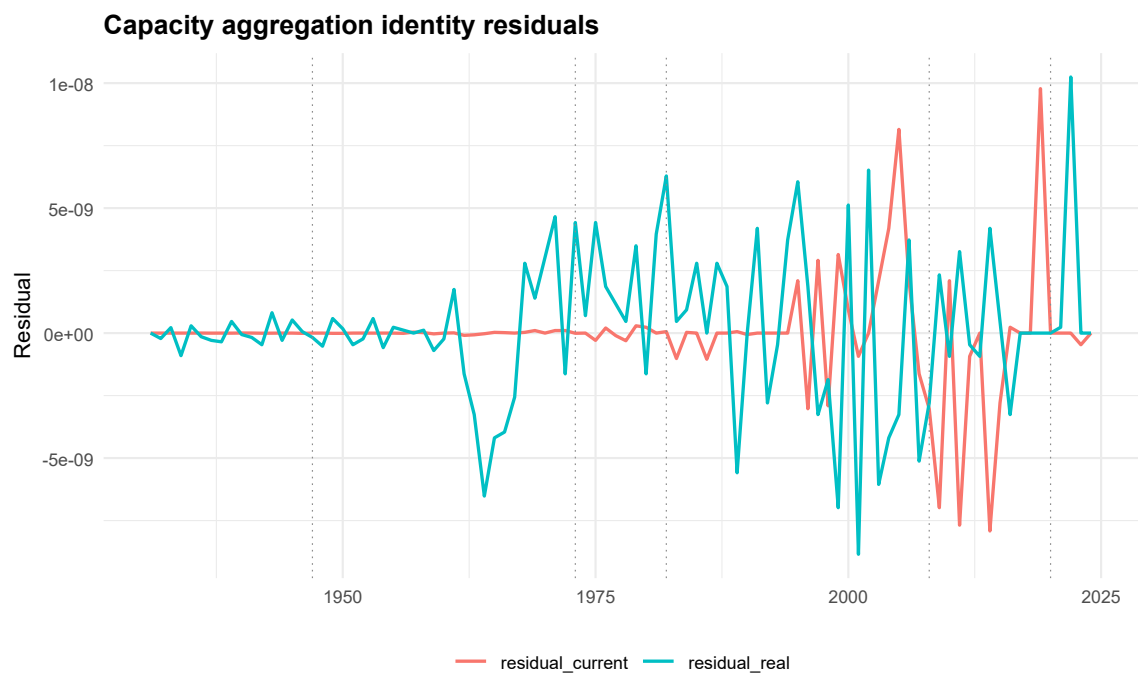


Figure 13: Capacity aggregation identity residuals.

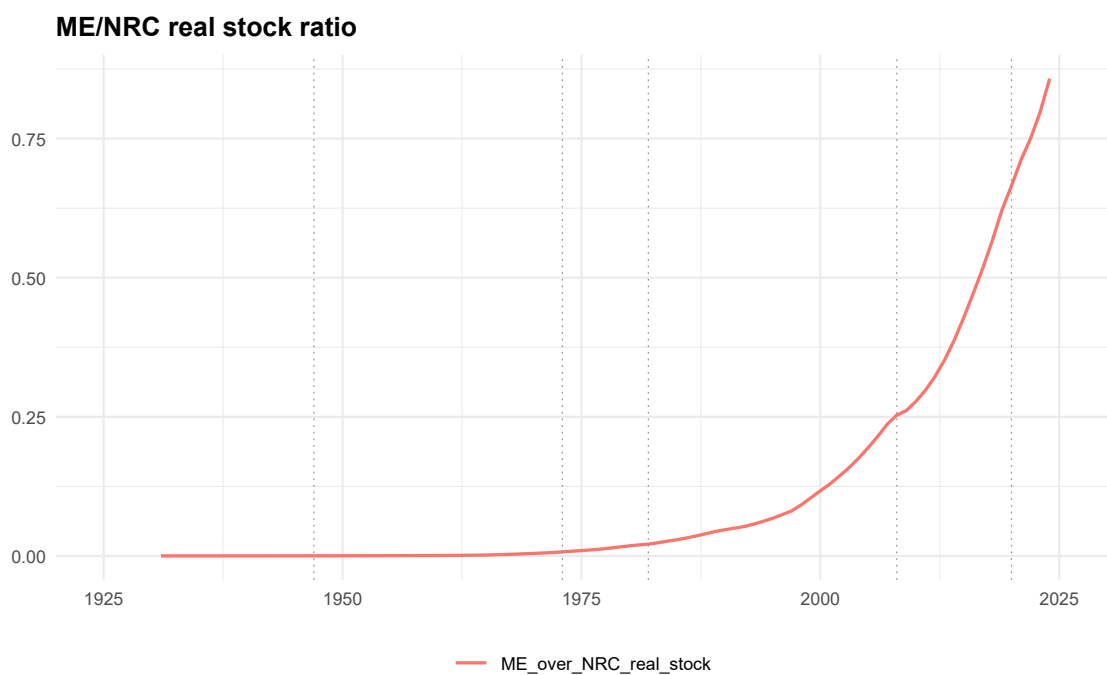


Figure 14: Figure 14 composition diagnostic.

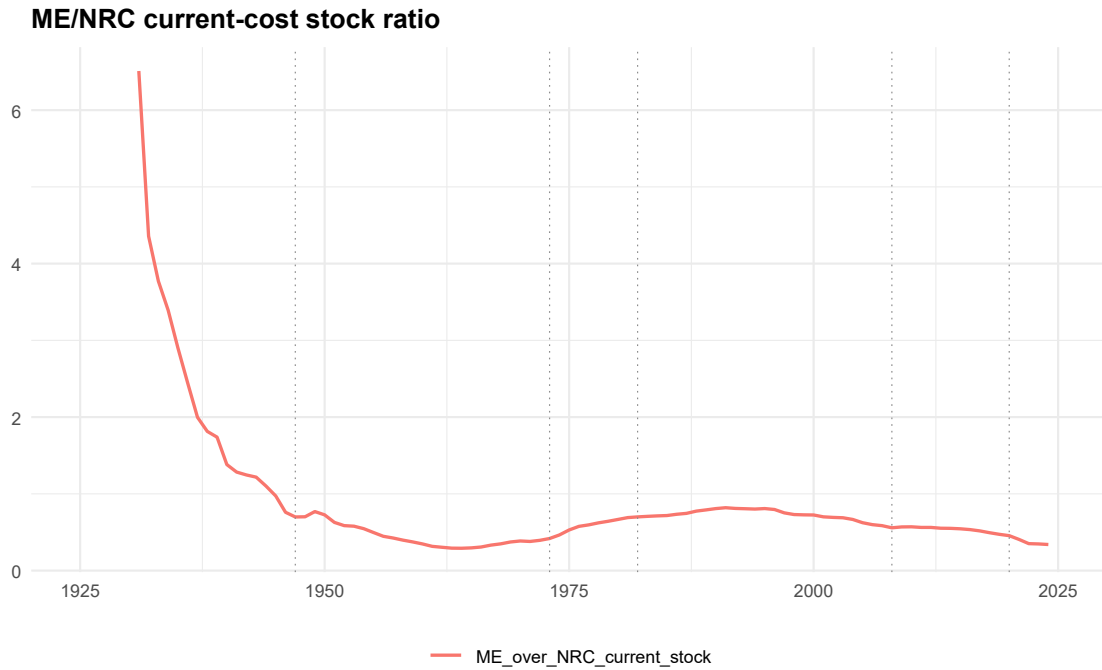


Figure 15: Figure 15 composition diagnostic.

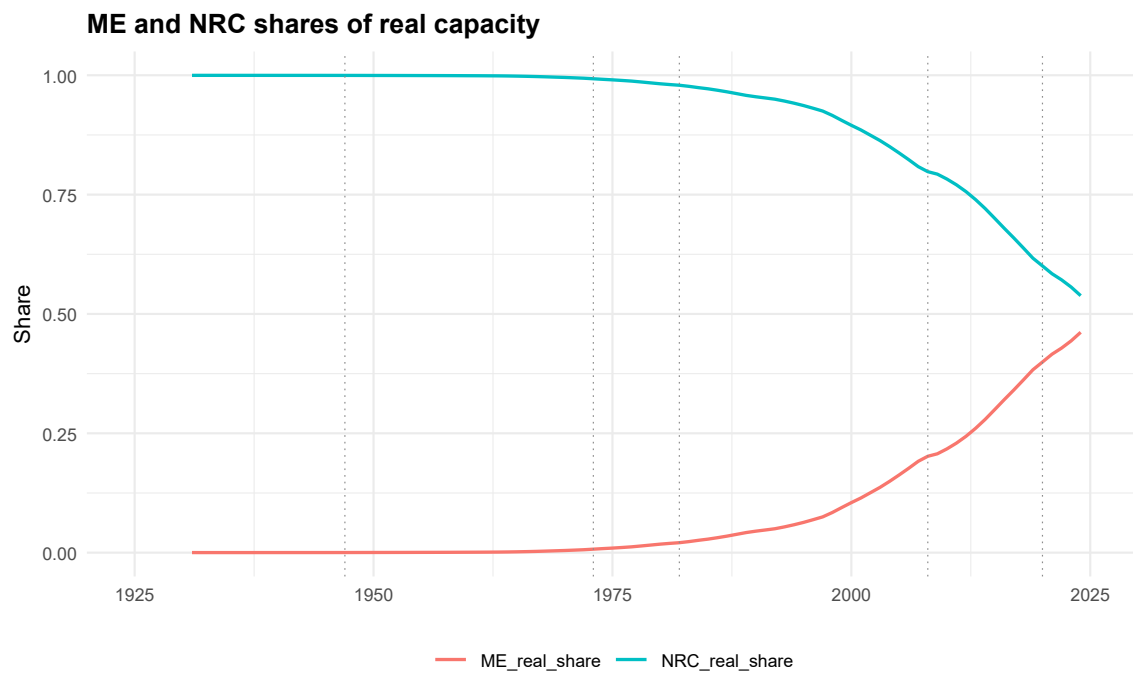


Figure 16: Figure 16 composition diagnostic.

11 Price and Valuation Diagnostics

12 Output-Capital Level Relations

13 Scatterplots of Locked Level Relations

14 Distribution and Surplus-Share Inspection

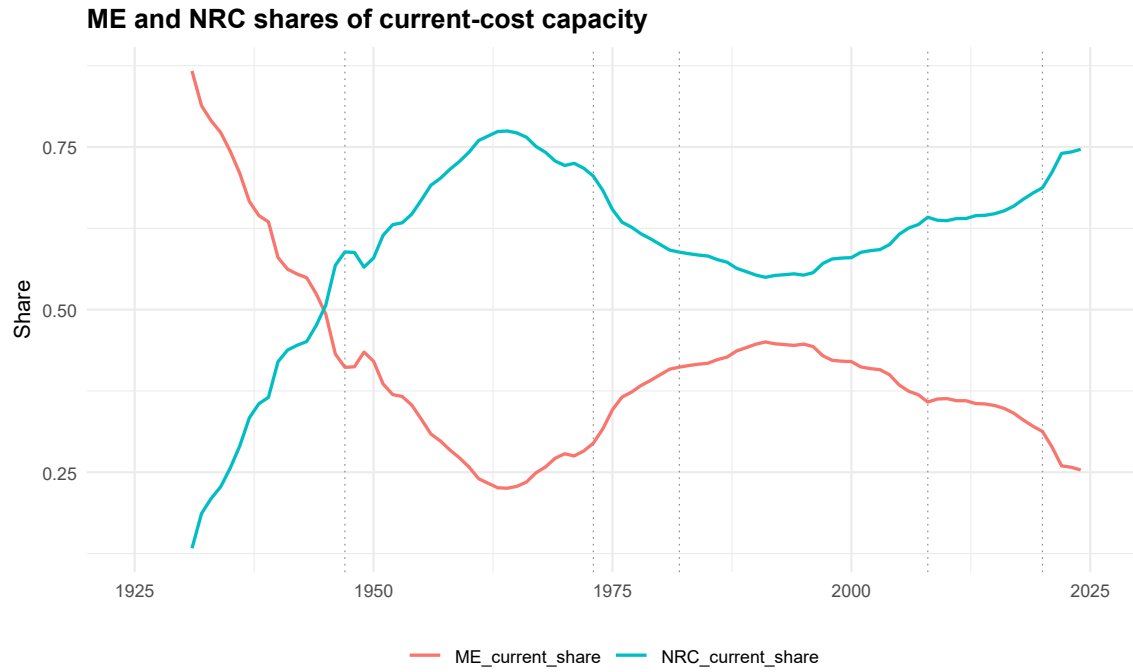


Figure 17: Figure 17 composition diagnostic.

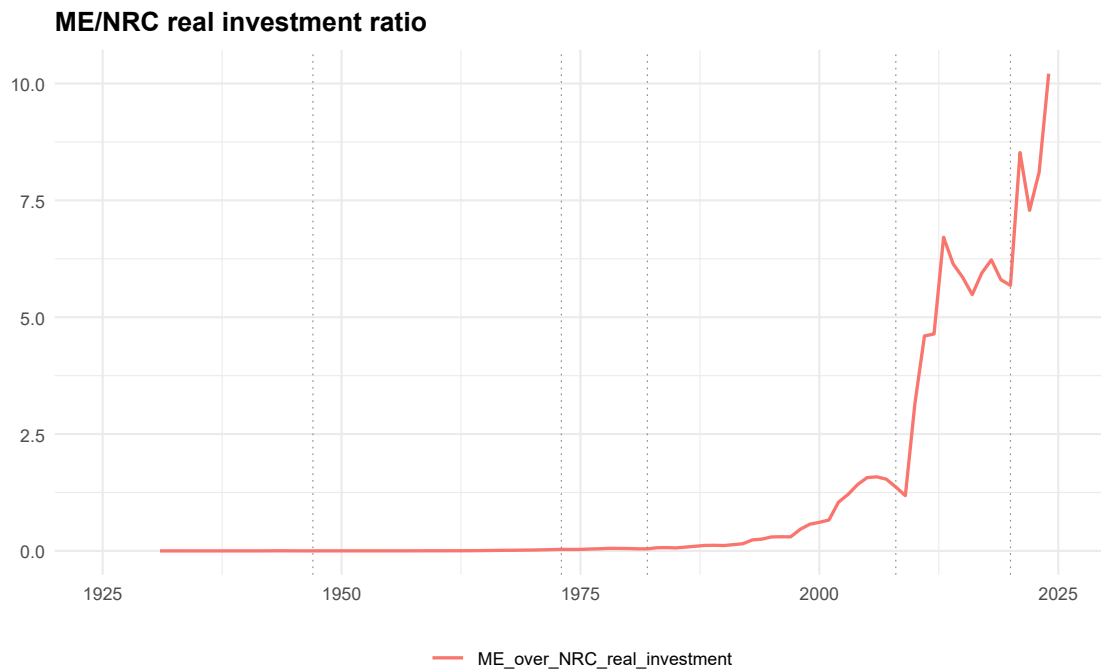


Figure 18: Figure 18 composition diagnostic.

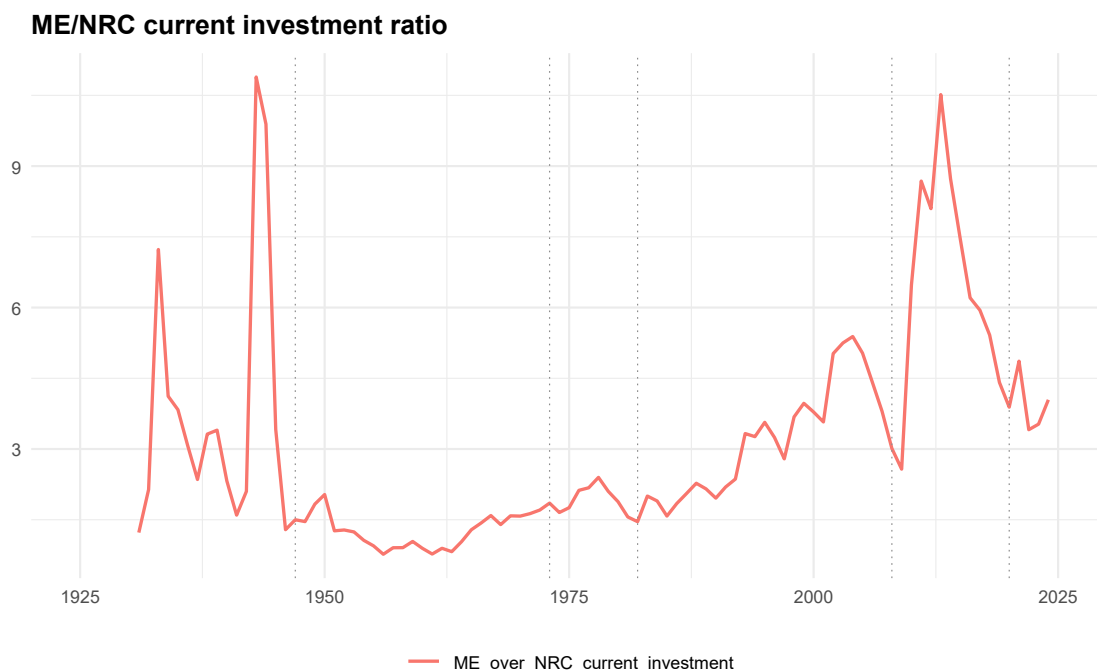


Figure 19: Figure 19 composition diagnostic.

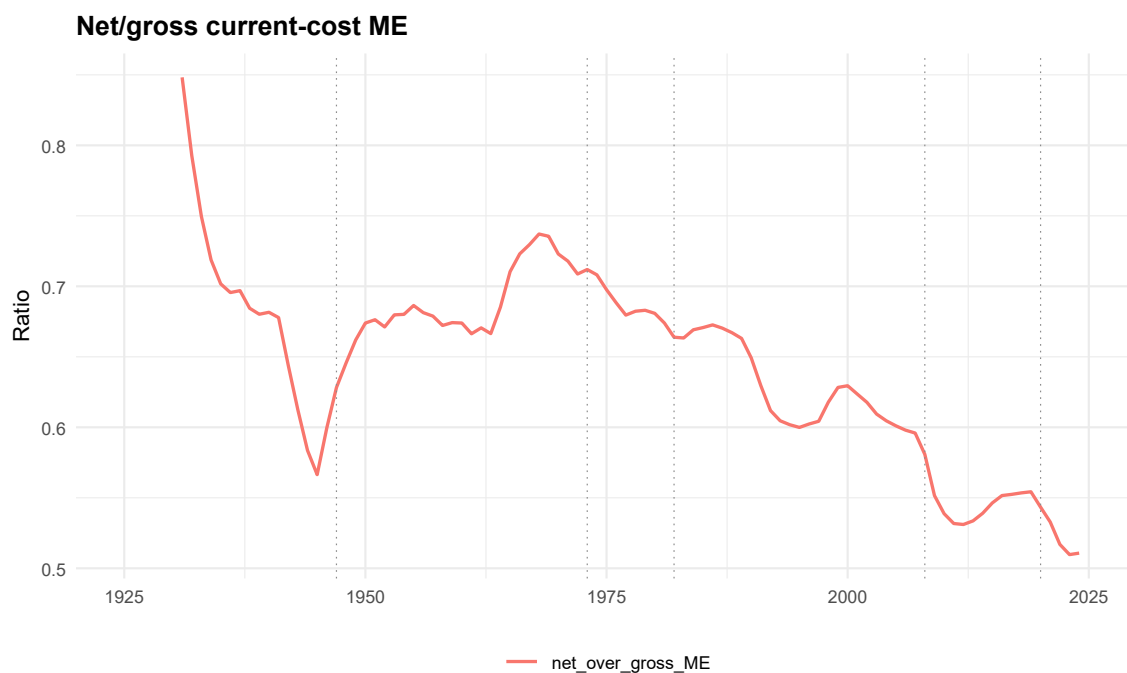


Figure 20: Net-over-gross current-cost ME comparison.

15 Financial-Sector Transfer and Double-Counting Safeguards

Financial-sector operating surplus and imputed-interest corrections are not silently treated as productive-origin surplus; they remain transfer, reconciliation, or candidate layers unless semanti-

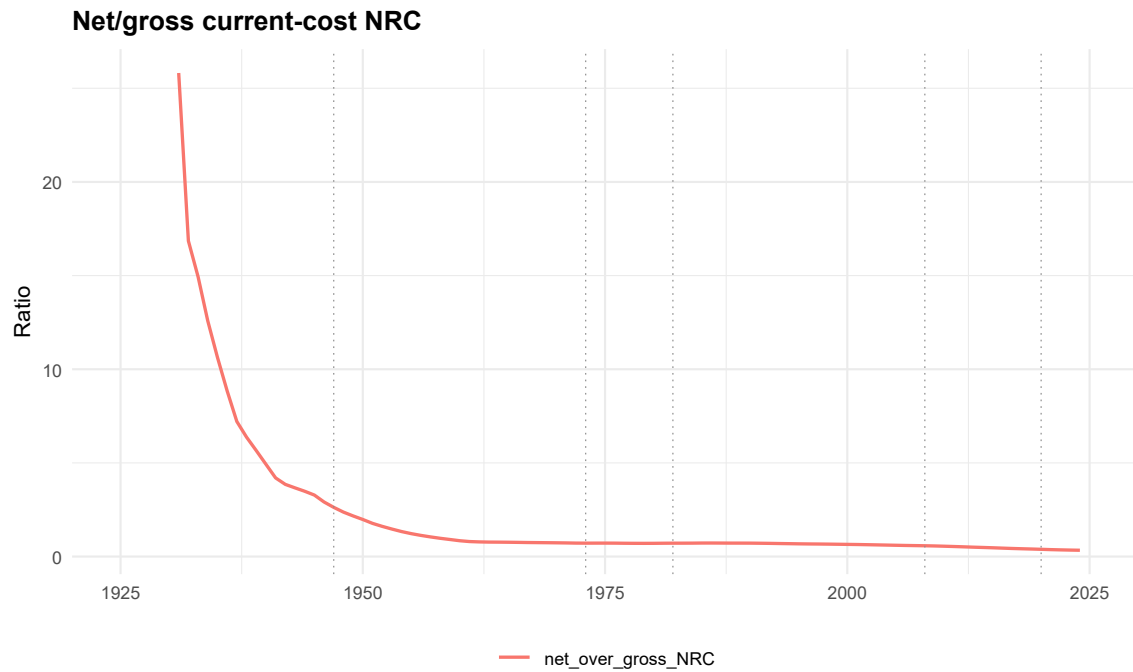


Figure 21: Net-over-gross current-cost NRC comparison.

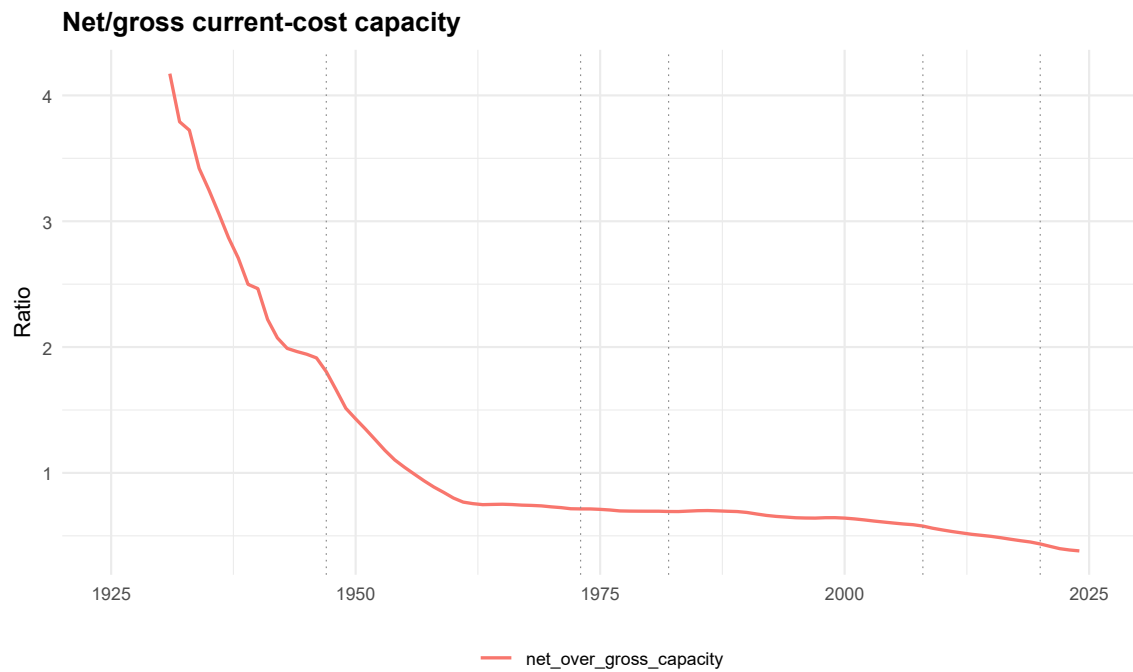


Figure 22: Net-over-gross current-cost capacity comparison.

cally validated.

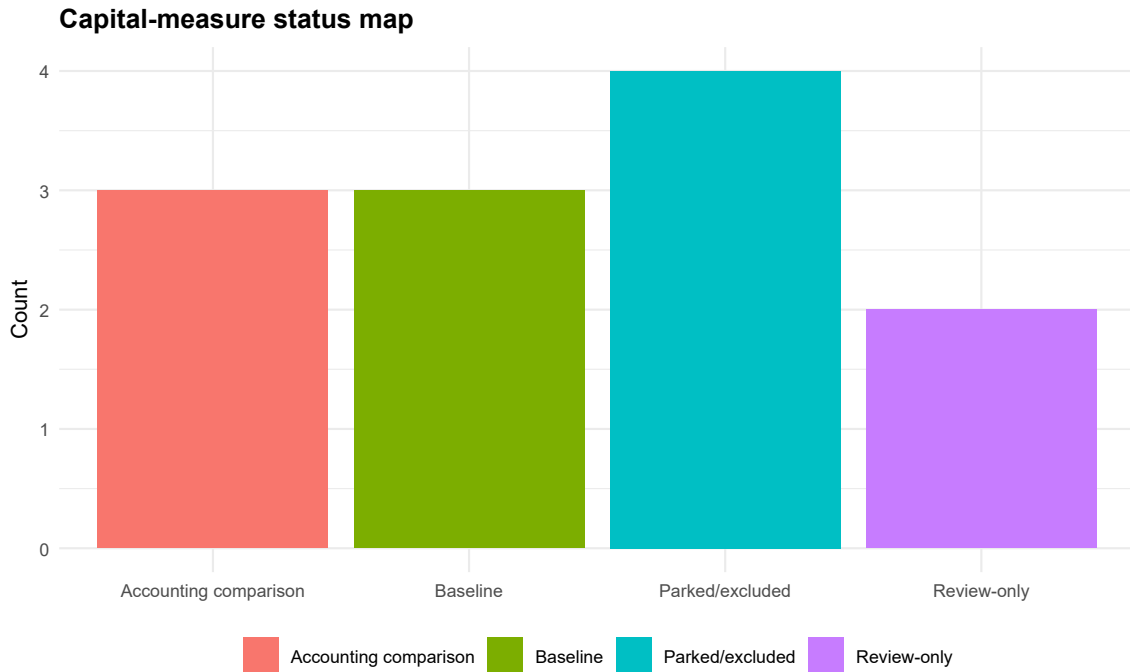


Figure 23: Capital-measure status map.

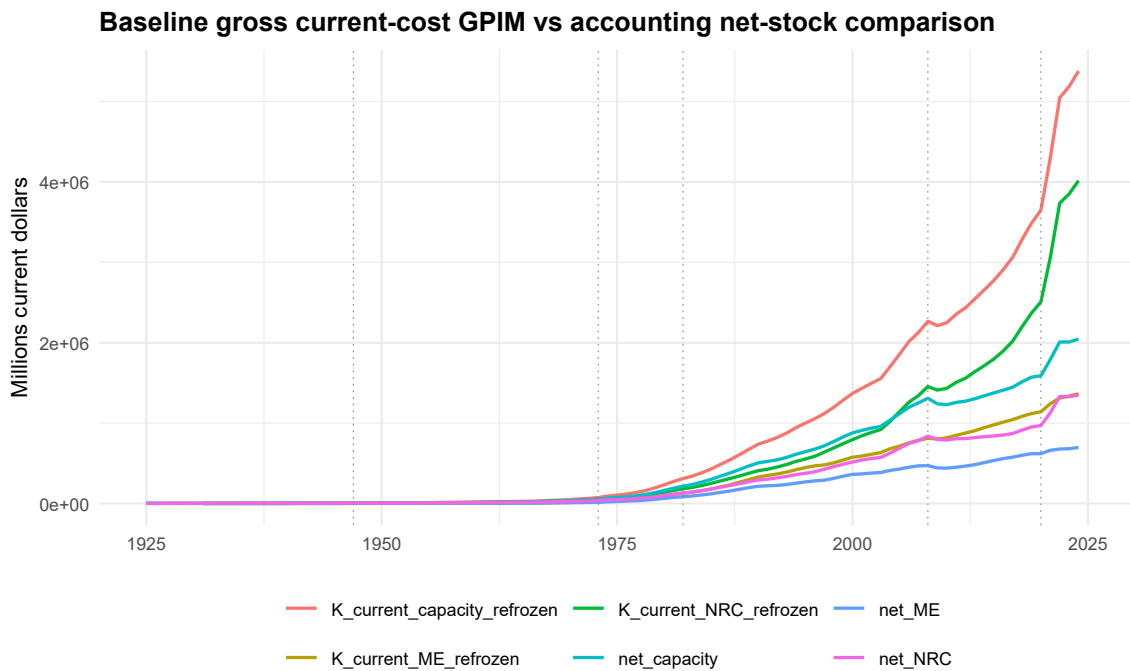


Figure 24: Baseline gross GPIM versus accounting net-stock comparison.

layer	status	safeguard
NFC productive-origin baseline	authorized baseline/accounting	productive-origin baseline
Corporate reconciliation variants	authorized reconciliation	comparison only
Financial transfer/correction candidates	candidate only	not silently productive-origin su
Imputed-interest candidates	candidate only	requires semantic crosswalk

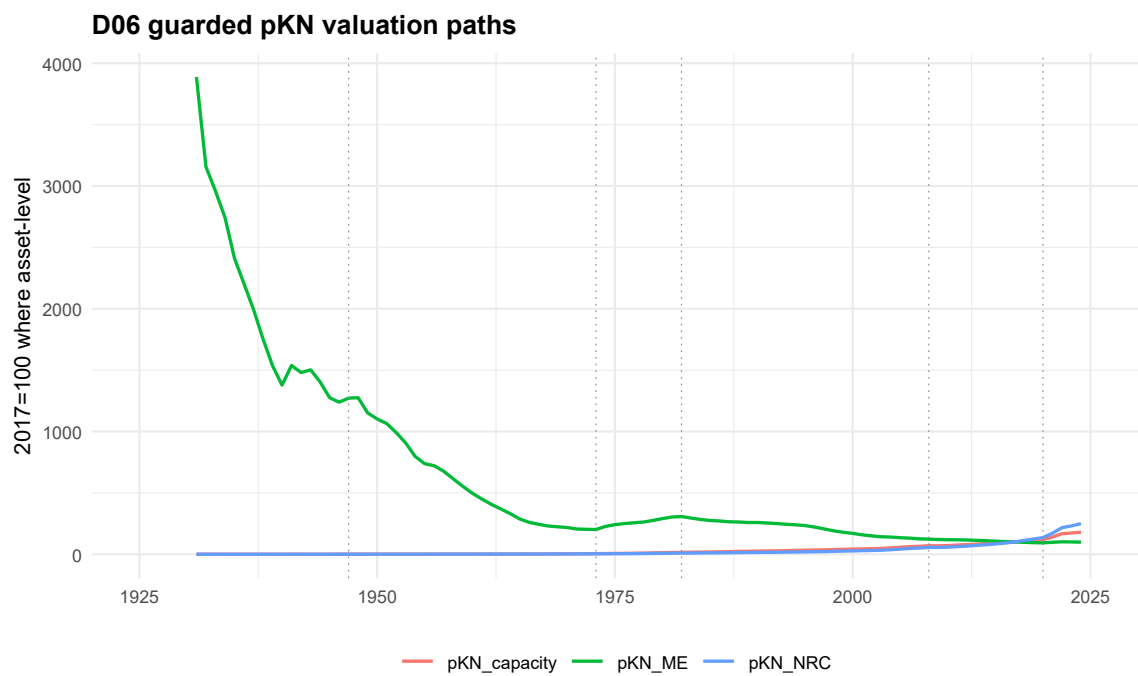


Figure 25: Guarded pKN valuation diagnostics.

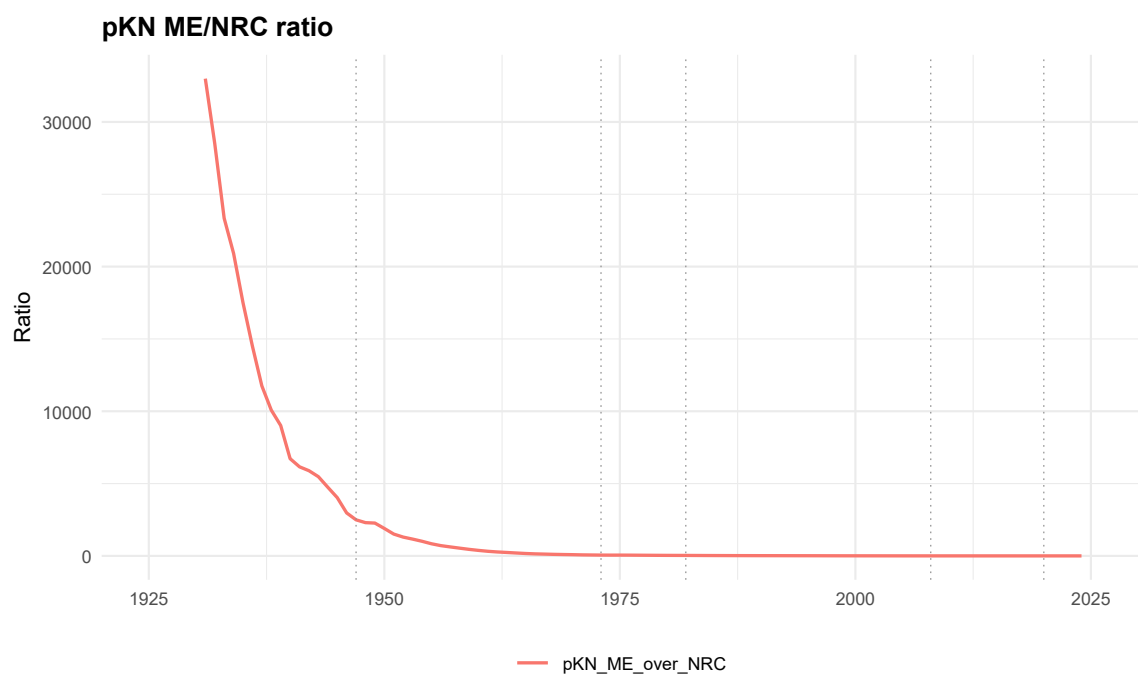


Figure 26: pKN ME/NRC ratio.

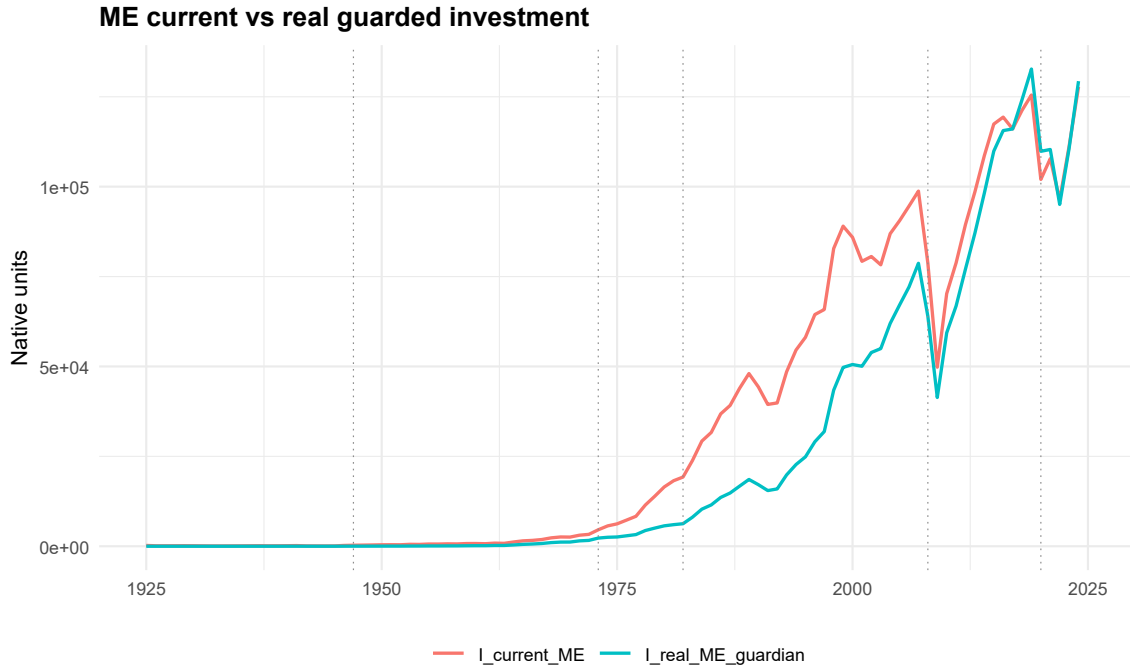


Figure 27: ME current versus real investment.

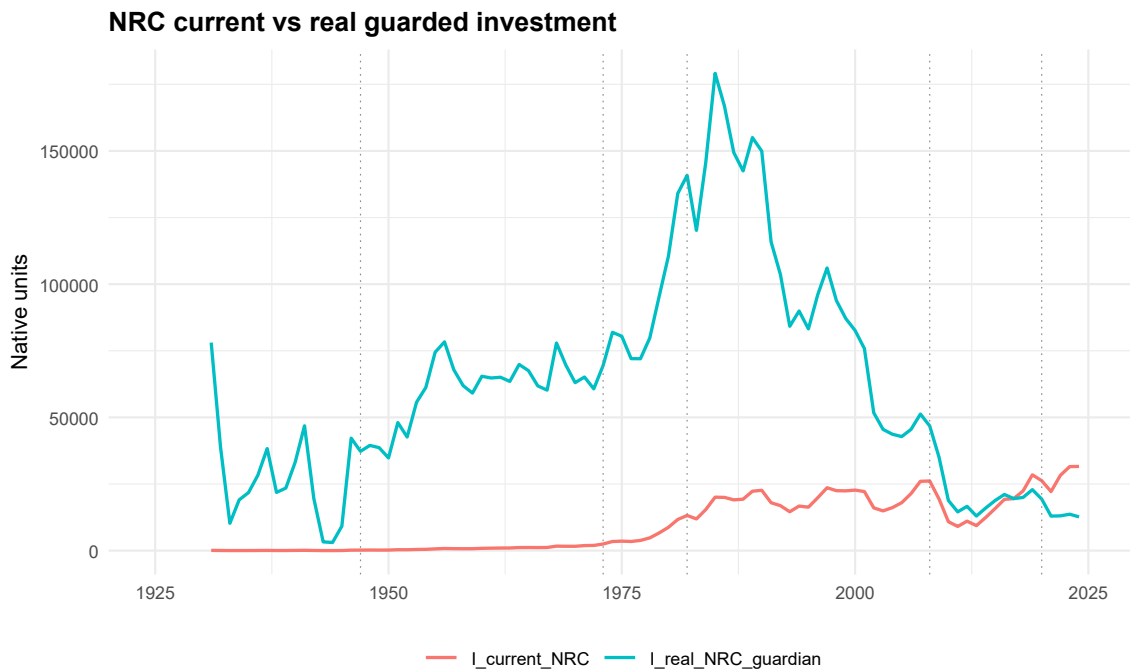


Figure 28: NRC current versus real investment.

16 Capital-Output-Distribution Relational Inspection

17 Historical Visual Annotations

Main time-series figures mark 1947, 1973, 1982, 2008, and 2020 as visual annotations only. No periodized model variable is created.

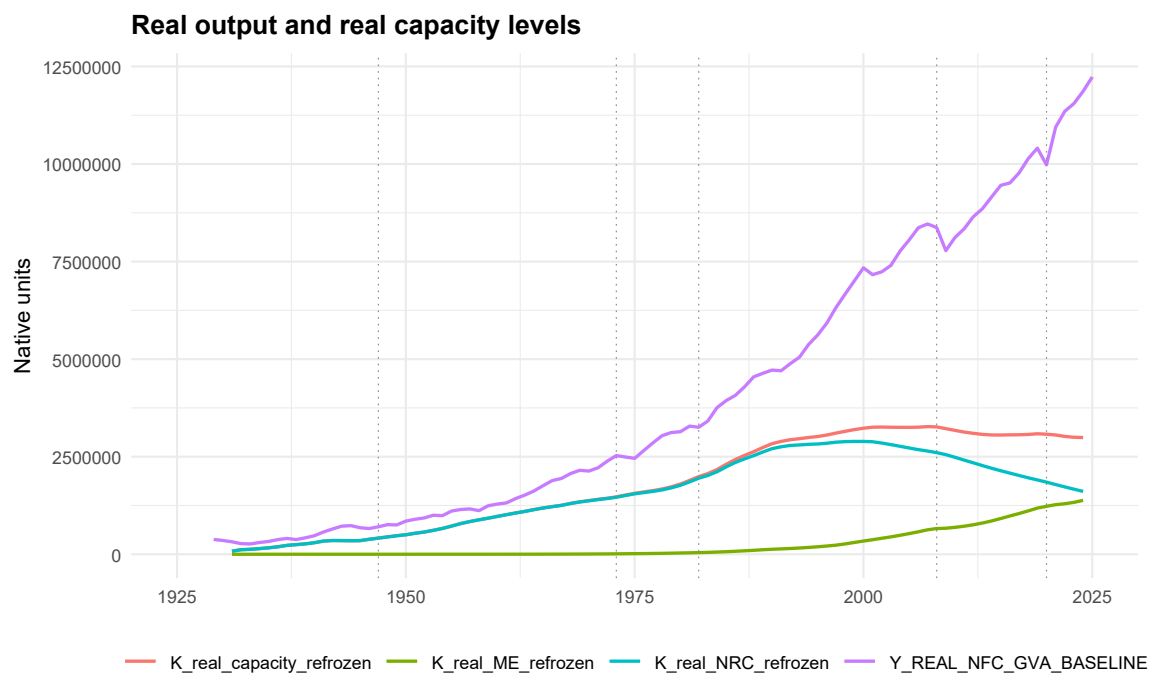


Figure 29: Figure 30 output-capital relation.

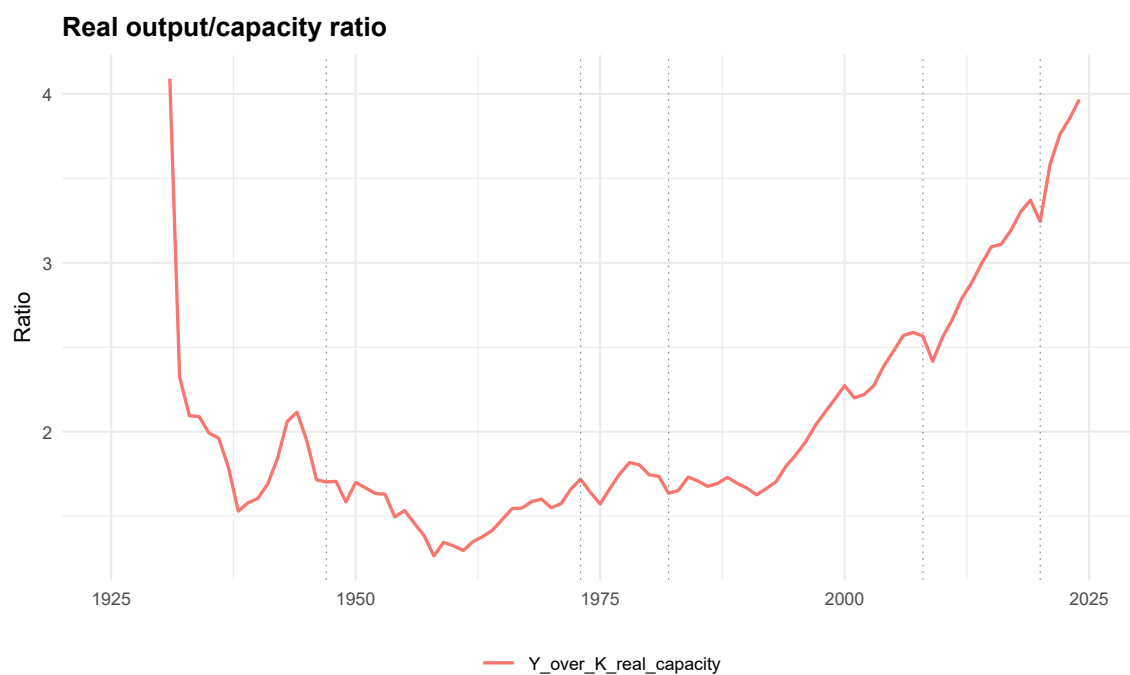


Figure 30: Figure 31 output-capital relation.

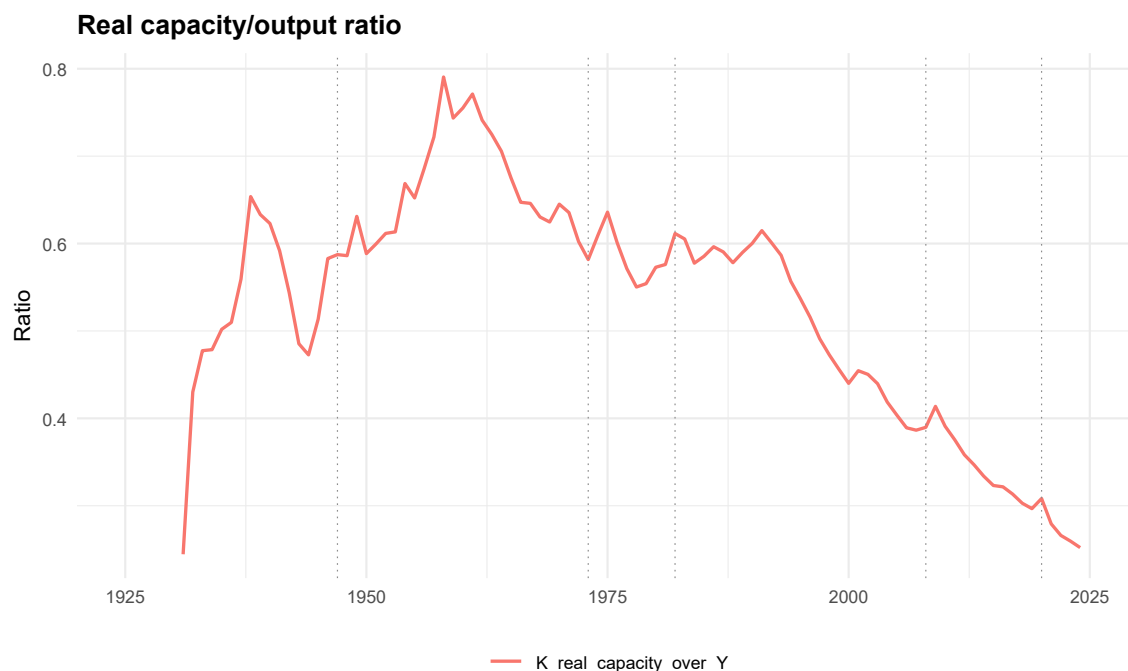


Figure 31: Figure 32 output-capital relation.

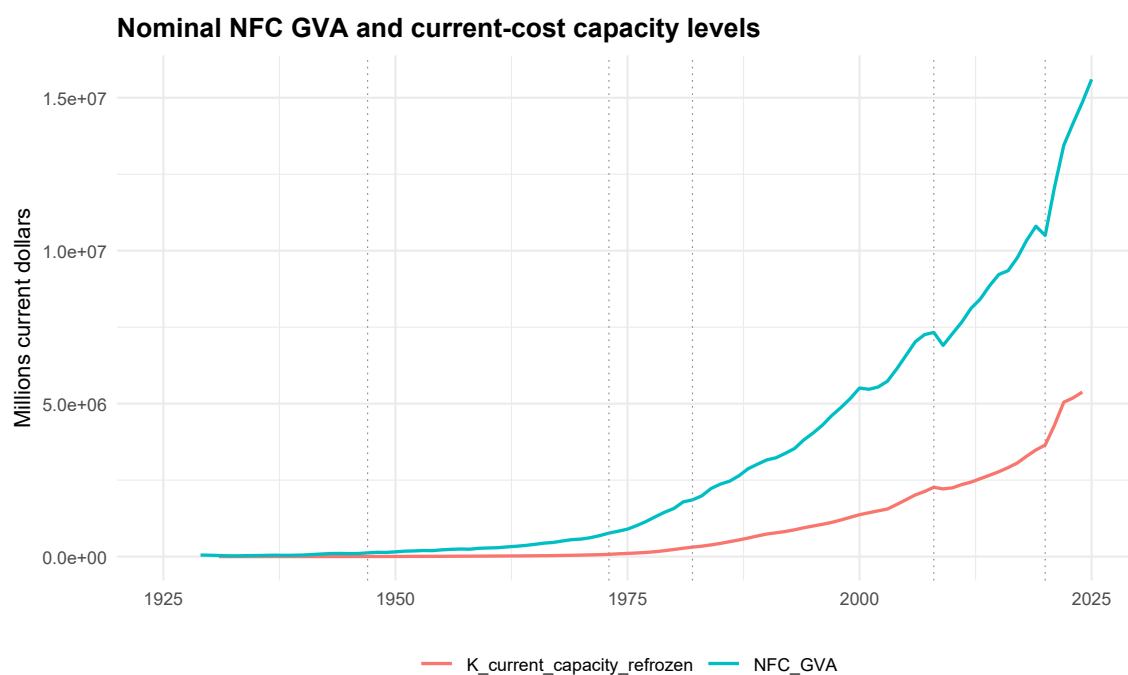


Figure 32: Figure 33 output-capital relation.

18 Human Review Checklist and Decision Gate

item	status	evidence
GPIM gross stock levels visually preserve accumulation tendency	PASS	Figures 1-8
ME/NRC composition visually plausible	PASS	Figures 14-
Stock-flow consistency residuals acceptable	PASS	Figure 11 a
Current-cost valuation identities acceptable	PASS	Figure 12 a
pKN/valuation diagnostics acceptable	PASS	Figures 26-

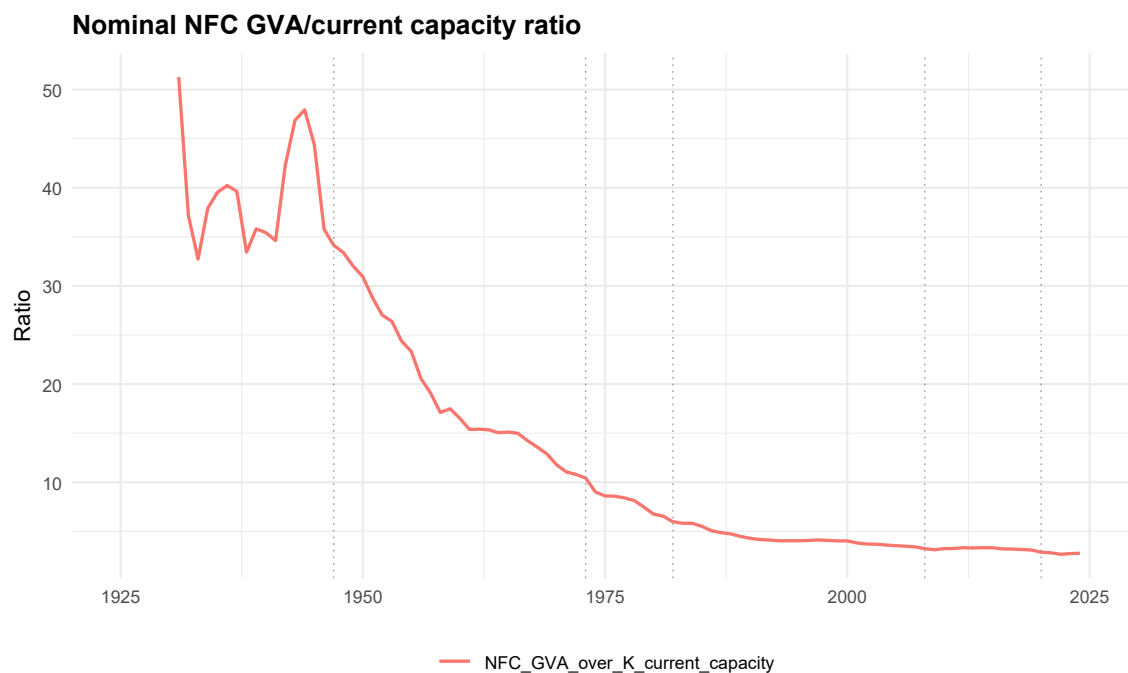


Figure 33: Figure 34 output-capital relation.

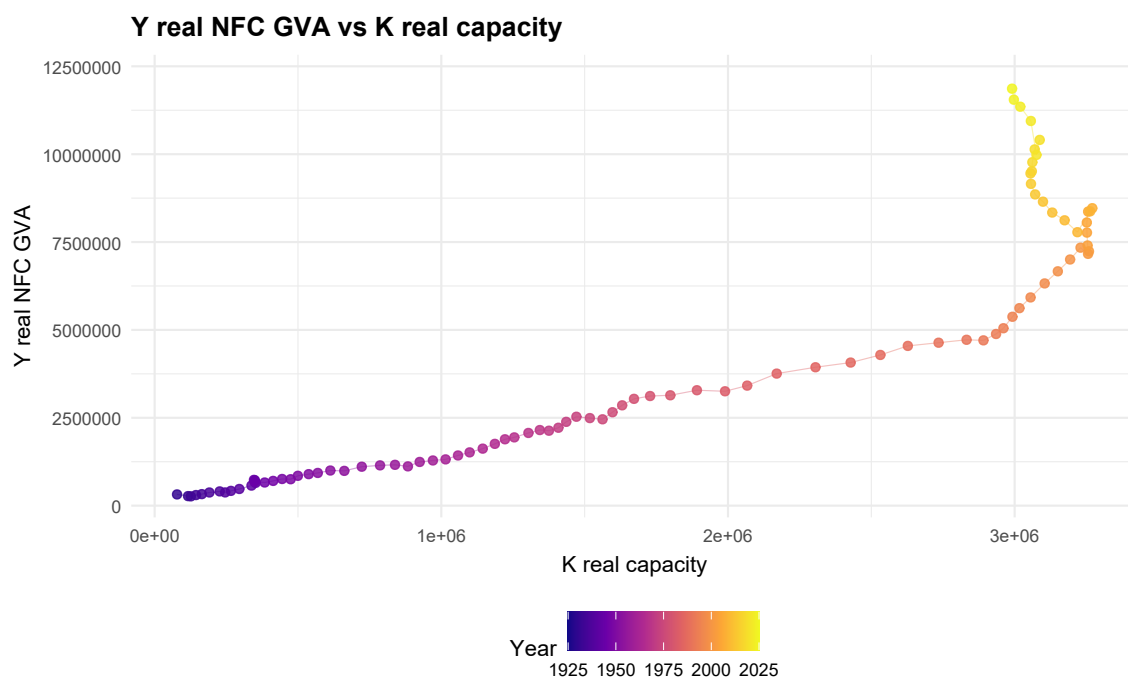


Figure 34: Y real NFC GVA versus real capacity.

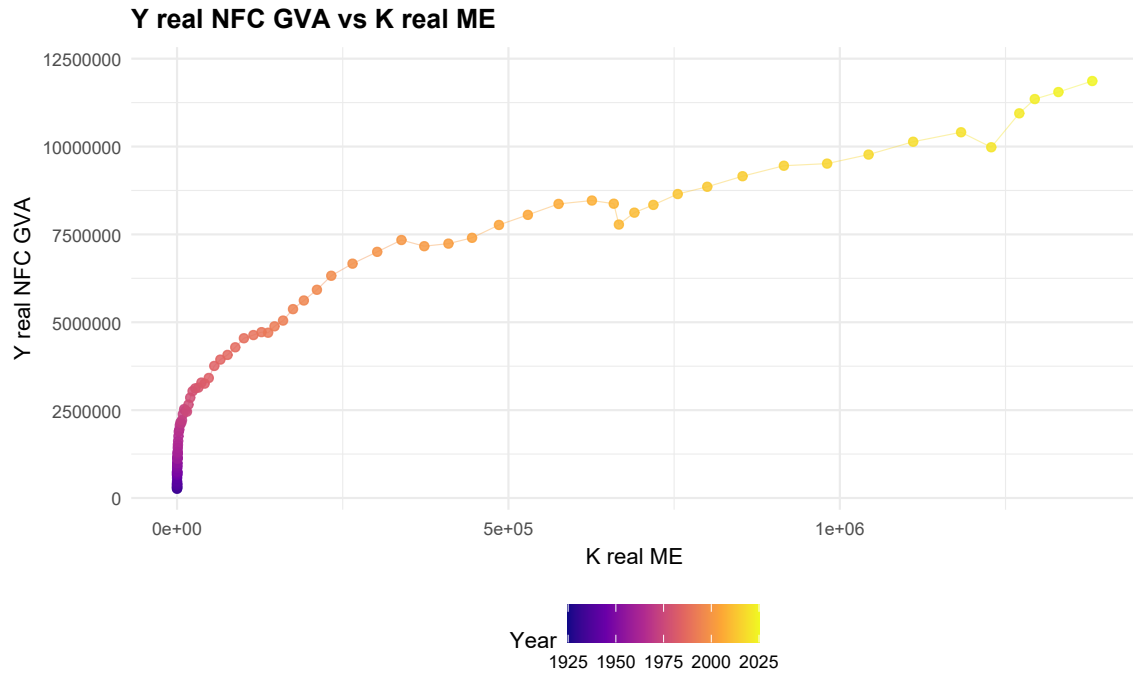


Figure 35: Y real NFC GVA versus real ME.

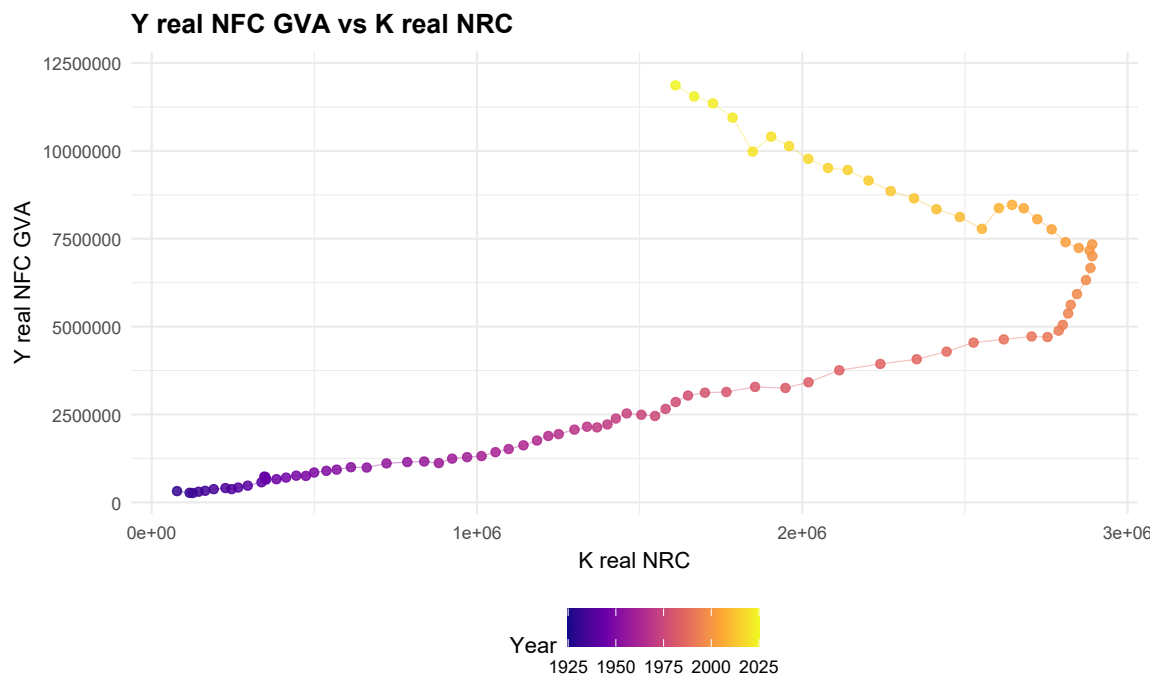


Figure 36: Y real NFC GVA versus real NRC.

A Secondary Figures

B Audit Tables

See the D09 CSV audit outputs and the tables included in the compilation bundle.

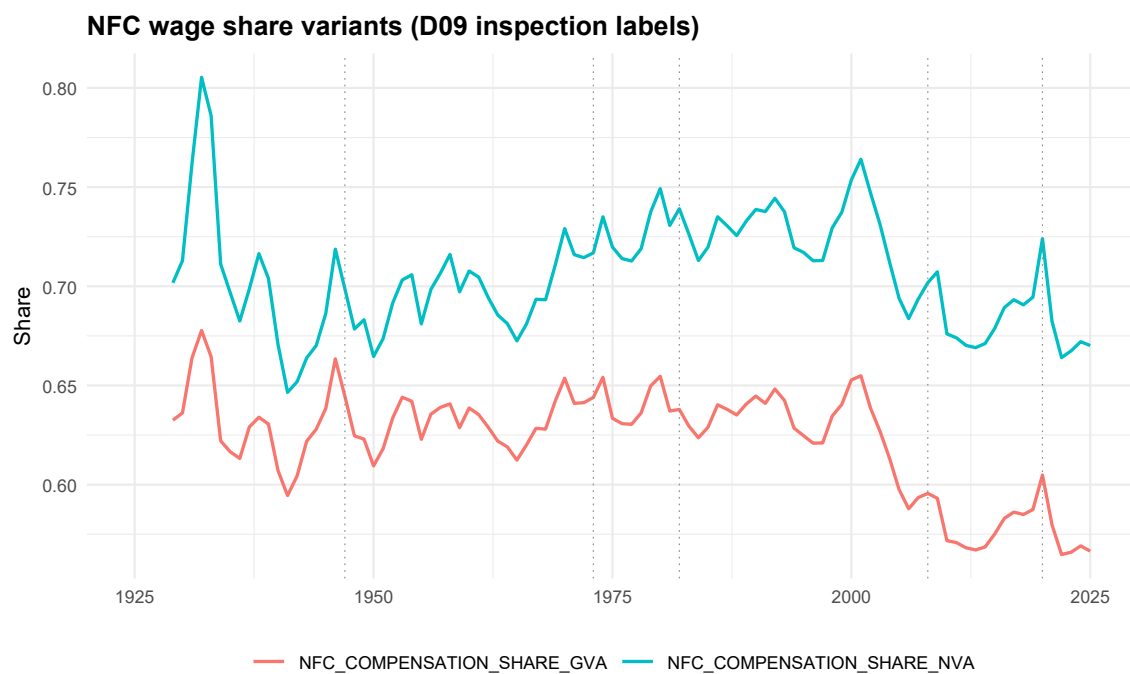


Figure 37: Figure 38 distribution diagnostic.

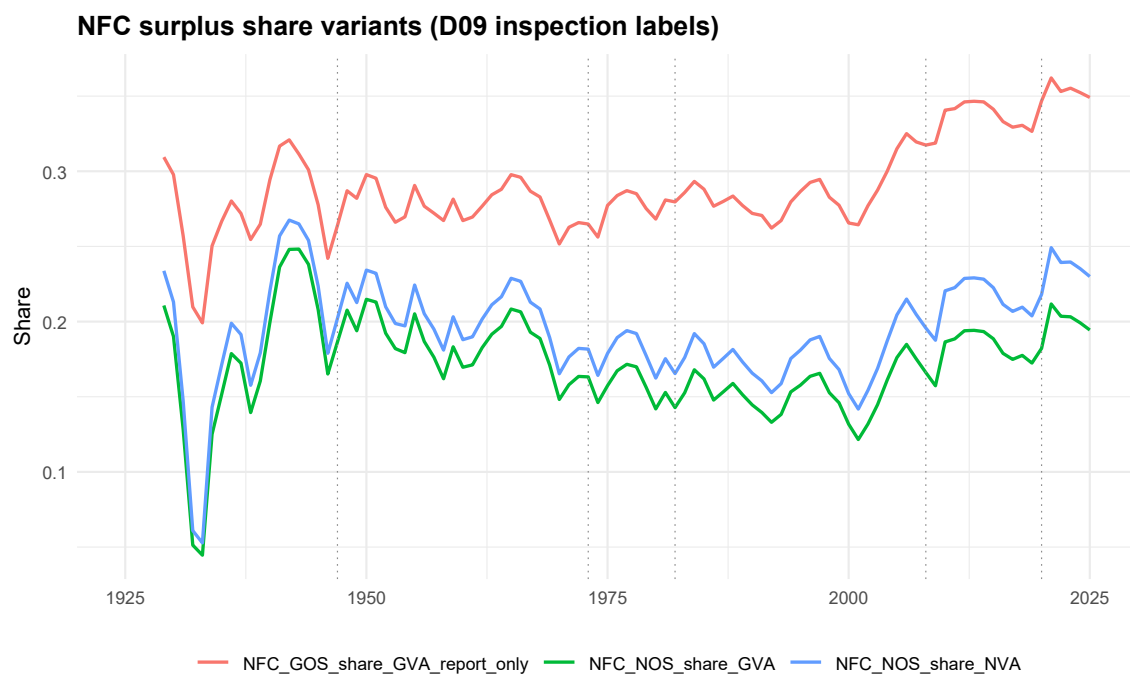


Figure 38: Figure 39 distribution diagnostic.

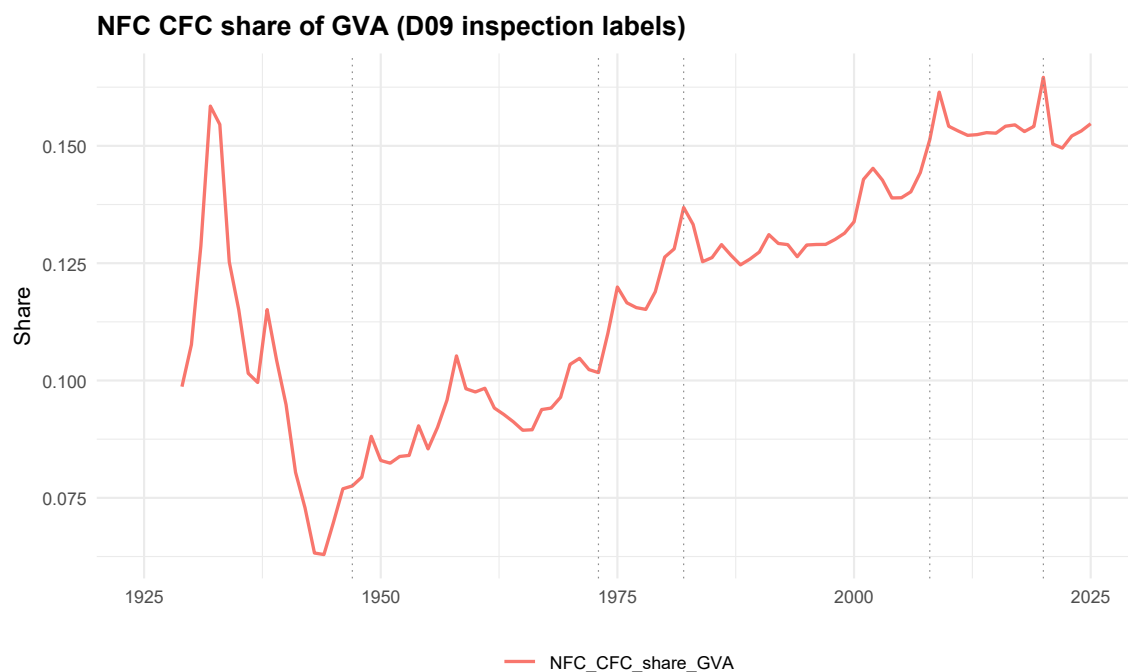


Figure 39: Figure 40 distribution diagnostic.

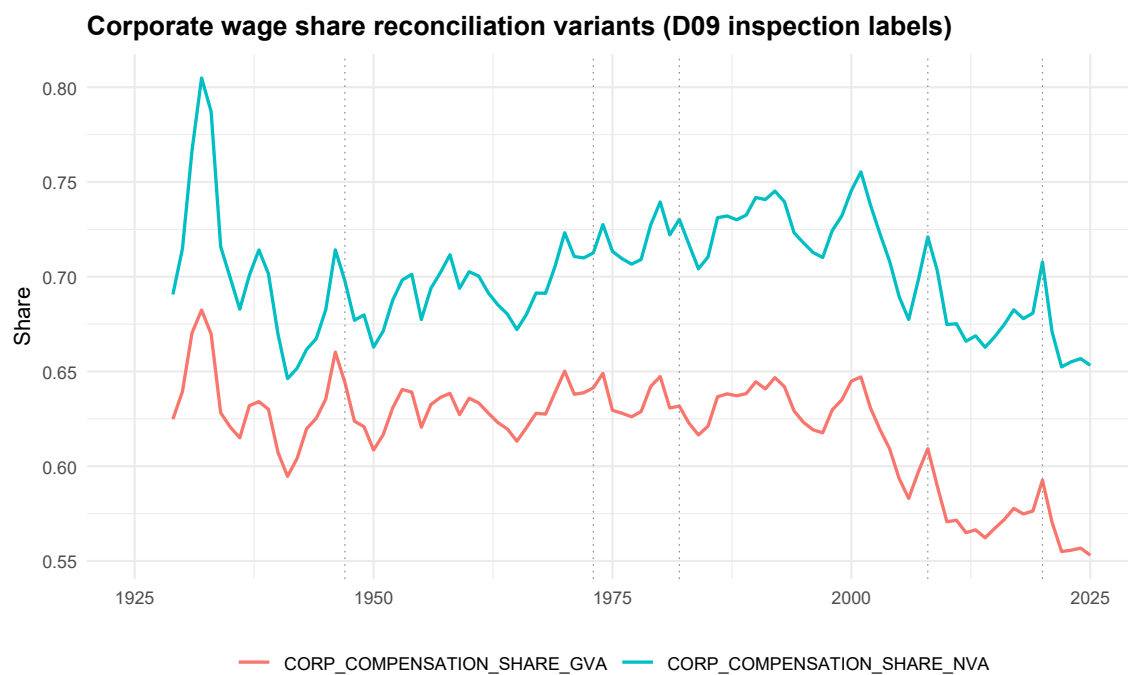


Figure 40: Figure 41 distribution diagnostic.

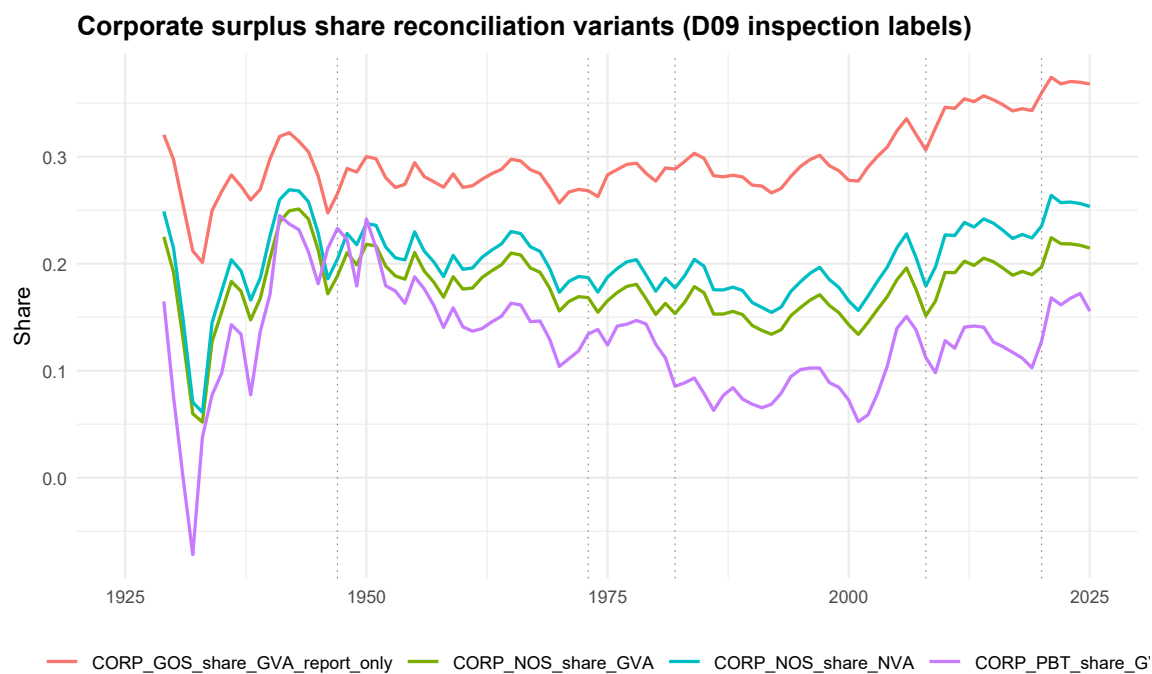


Figure 41: Figure 42 distribution diagnostic.

Surplus accounting ladder and double-counting safeguard



a authorized baseline/accounting
 a authorized reconciliation
 a candidate only

Figure 42: Surplus accounting ladder and double-counting safeguard.

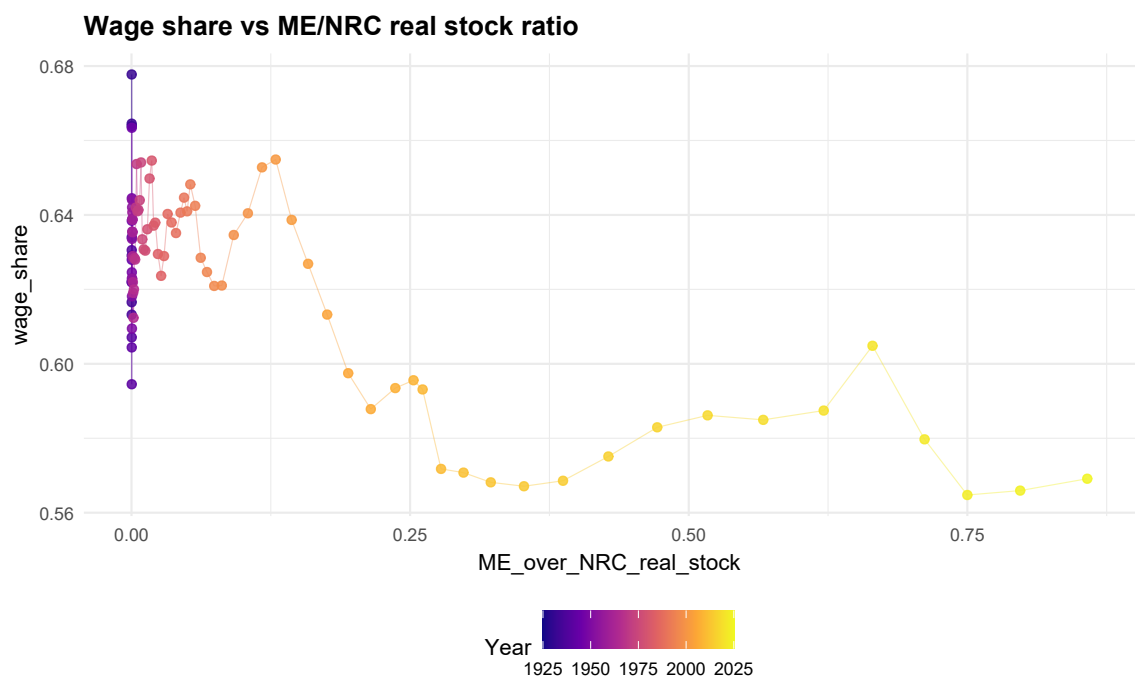


Figure 43: Figure 44 capital-output-distribution relation.

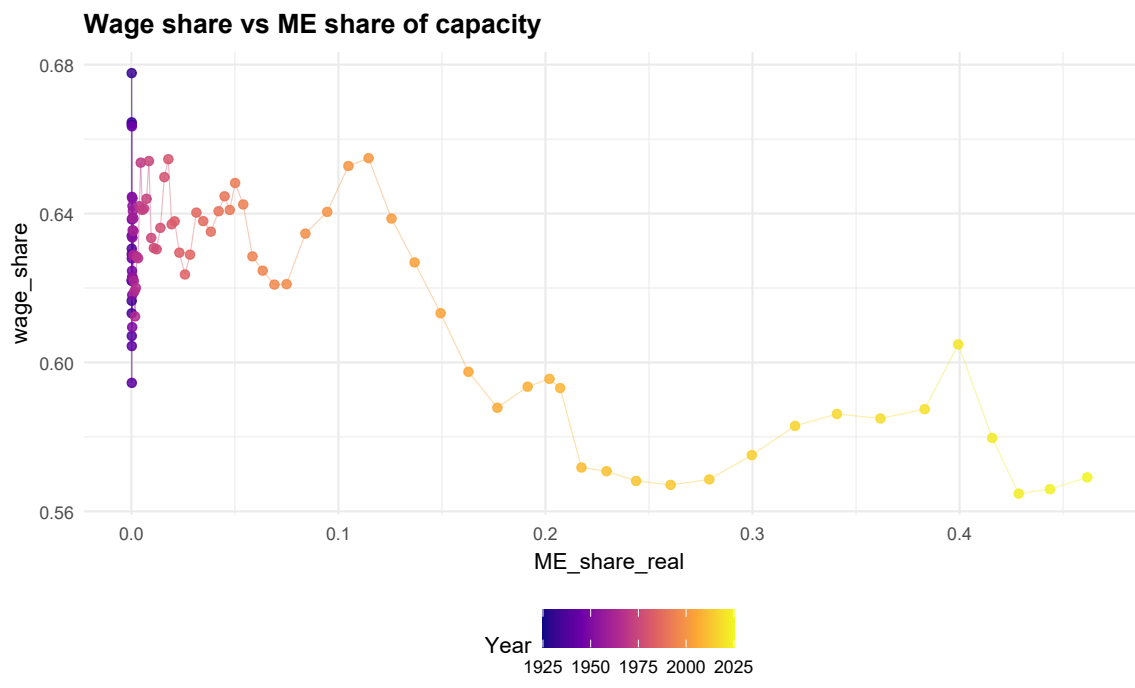


Figure 44: Figure 45 capital-output-distribution relation.

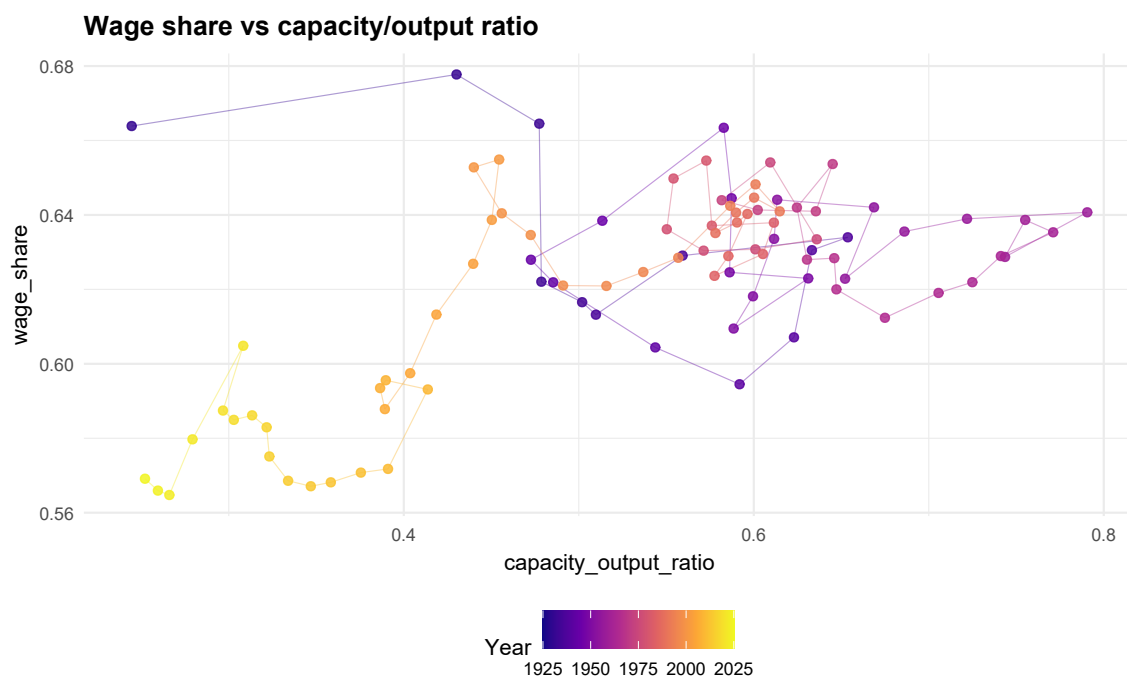


Figure 45: Figure 46 capital-output-distribution relation.

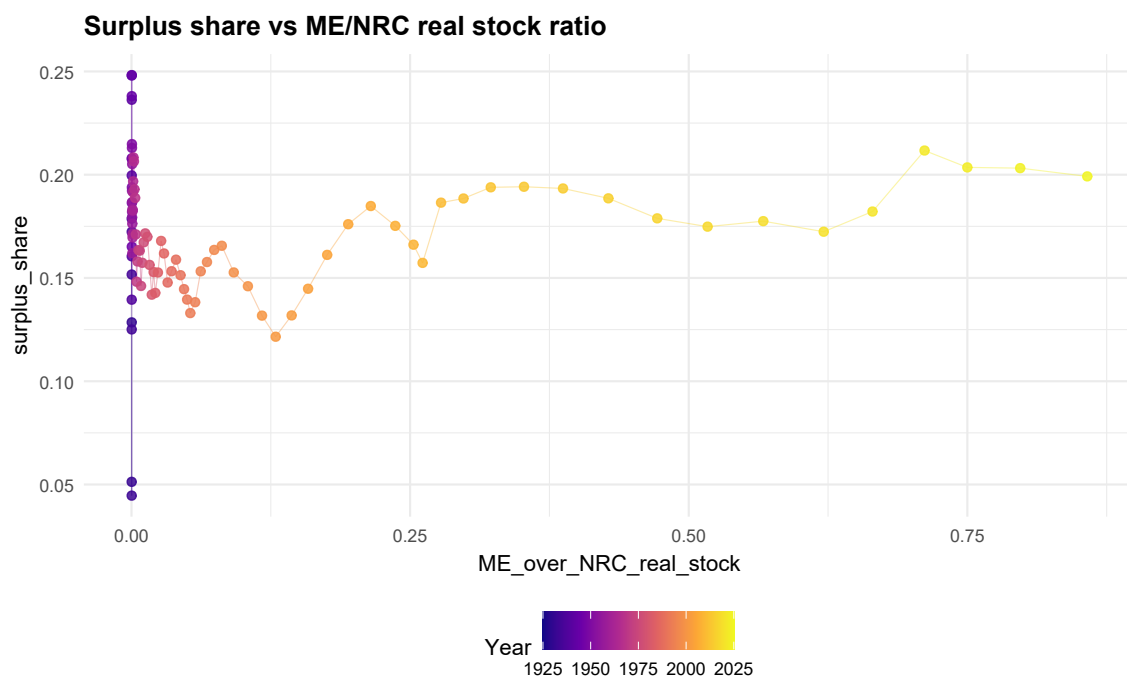


Figure 46: Figure 47 capital-output-distribution relation.

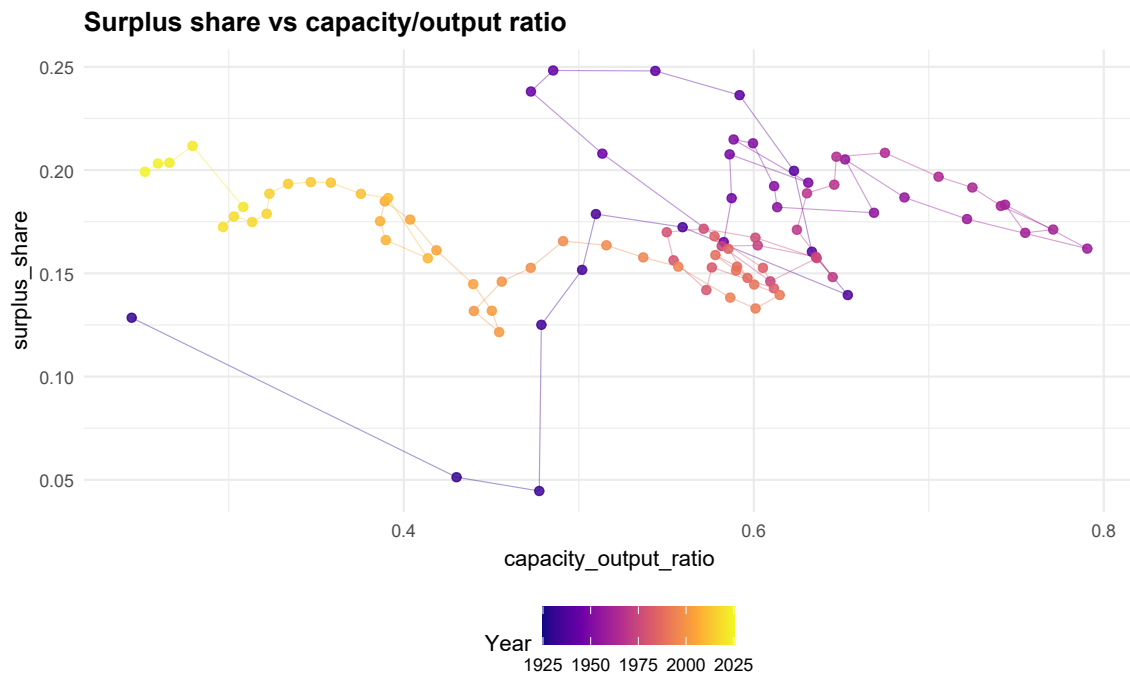


Figure 47: Figure 48 capital-output-distribution relation.

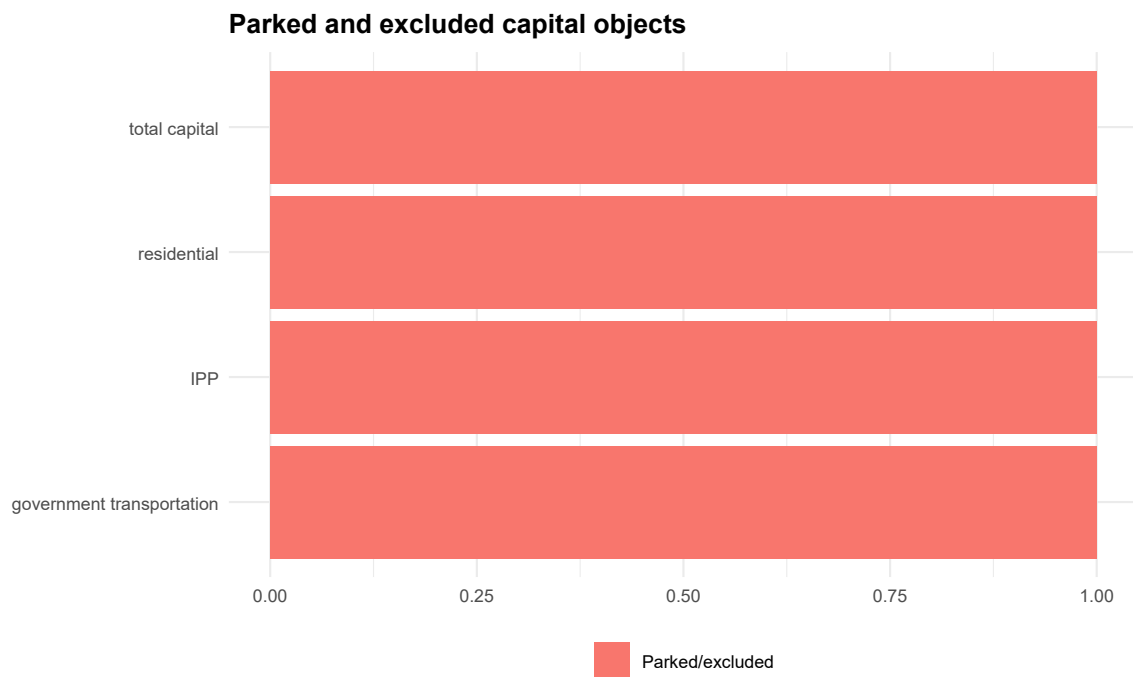


Figure 48: Parked and excluded capital objects, appendix status view.